



University of Frontier Technology, Bangladesh

Department of Internet of Things and Robotics Engineering

NirmalNode

Green IoT and Adaptive AI-Assisted Local Air Purification and Pollution Prediction System for Industrial Hotspots in Bangladesh

A thesis submitted in partial fulfilment of the requirements for the
degree of

Bachelor of Science in Internet of Things and Robotics Engineering

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Declaration

I hereby declare that the thesis titled “**NirmalNode: Green IoT and Adaptive AI-Assisted Local Air Purification and Pollution Prediction System for Industrial Hotspots in Bangladesh**” is an original work carried out by us under the supervision of Md. Ashiqussalehin, Lecturer, Department of Internet of Things and Robotics Engineering, University of Frontier Technology, Bangladesh, submitted in partial fulfilment of the requirements for the degree of Bachelor of Science in Internet of Things and Robotics Engineering. The work presented here has not been submitted elsewhere for the award of any other degree or diploma. All sources of information used have been duly acknowledged through appropriate citations.

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Approval

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Abstract

Rapid and largely unregulated industrialization in Bangladesh has created severe, spatially concentrated air pollution hotspots inside factories and workshops – welding bays, boiler rooms, generator rooms, chemical storage areas, and factory entrances – where workers are exposed to hazardous concentrations of particulate matter (PM_{1.0}, PM_{2.5}, PM₁₀), carbon monoxide (CO), nitrogen dioxide (NO₂), volatile organic compounds (VOC), ammonia (NH₃), hydrogen sulphide (H₂S), and smoke for prolonged periods. Conventional air quality management strategies in this context are almost exclusively city-scale or plant-scale, relying on either large centralized purification plants or coarse regulatory monitoring, neither of which protects a worker standing directly beside a hazardous point source. This thesis proposes **NirmalNode**, a Green Internet of Things (IoT) and Adaptive Artificial Intelligence (AI)-assisted system that performs *local, hotspot-level* air purification rather than area-wide purification, activating a compact filtration unit only at the specific micro-location where a worker is exposed, and only when it is needed.

NirmalNode is built around a low-cost ESP32 microcontroller node instrumented with a PMS5003 particulate sensor, MiCS-4514 and MQ-series gas sensors, and an optional BME280 environmental sensor, which continuously streams pollutant, temperature, humidity, and pressure readings, tagged with time, day, and location identifiers, to a cloud database over Wi-Fi. A lightweight, edge-deployable Adaptive AI model – trained initially on historical Bangladesh air-quality data and subsequently updated using *incremental (online) learning* on live sensor streams – forecasts pollutant concentrations one to two hours ahead of the current time. When the forecast crosses a hazard threshold, the system pre-emptively activates a DC fan that draws air through a cyclone separator, a HEPA filter, and an activated-carbon filter stage before releasing purified air back into the worker’s immediate breathing zone, so that purification begins *before* pollution becomes dangerous rather than reacting after exposure has already occurred.

The central methodological contribution of this work is the use of incremental/online learning models (including SGD Regressor, Passive-Aggressive Regressor, Hoeffding Tree, and Adaptive Random Forest, implemented using the River and scikit-learn libraries) in place of periodic full retraining. This allows a single deployed unit to gradually adapt its internal pollution model to a new site’s characteristics – for example, when a unit is relocated from an industrial cluster in Gazipur to one in Narayanganj – without requiring an engineer to manually retrain and redeploy firmware, and while consuming a fraction of the computational and energy budget of a conventional deep-learning retraining cycle. This area-specific adaptivity, combined with the deliberately lightweight (“Green AI”) model choice and the low-power ESP32-centred hardware

(“Green IoT”), is intended to make the system economically and technically feasible for widespread deployment across the many small and medium industrial units that characterize Bangladesh’s manufacturing sector.

A bench-scale prototype of NirmalNode was designed, instrumented, and evaluated in a controlled laboratory setting and, where feasible, at a representative industrial site, with performance assessed along three axes: (i) predictive accuracy of the adaptive pollution forecasting model, measured using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Precision, Recall, F1-score, and coefficient of determination (R^2); (ii) purification efficiency of the cyclone–HEPA–activated-carbon filter train, measured as percentage reduction in PM and gaseous pollutant concentration; and (iii) system-level power consumption and estimated deployment cost, to establish the practical viability of the design for resource-constrained industrial settings.

The findings and design decisions of this work are discussed against the backdrop of the existing literature on industrial air monitoring, indoor and portable air purification, and AI/IoT-based air quality prediction reviewed in Chapter 2, and the identified research gap – the absence of a low-cost, adaptive, hotspot-targeted purification system suited to the Bangladeshi industrial context – is used to motivate the specific contributions of NirmalNode. The thesis concludes with a discussion of the limitations of the present prototype and outlines a future work agenda spanning multi-node factory-wide deployment, federated learning across nodes, further edge-AI optimization, and a solar-powered variant for use in areas with unreliable grid power.

Keywords: Green IoT; Green AI; Adaptive AI; Incremental Learning; Air Pollution Prediction; Local Air Purification; ESP32; HEPA Filter; Activated Carbon; Cyclone Separator; Industrial Hotspot Monitoring; Edge–Cloud Architecture; Bangladesh.

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List of Abbreviations

AI	Artificial Intelligence
API	Application Programming Interface
ARF	Adaptive Random Forest
BME280	Bosch Environmental Sensor (Pressure, Humidity, Temperature)
CO	Carbon Monoxide
DHT	Digital Humidity and Temperature Sensor
ESP32	Espressif 32-bit System-on-Chip Microcontroller
GPRS	General Packet Radio Service
H₂S	Hydrogen Sulphide
HEPA	High-Efficiency Particulate Air (filter)
IAQ	Indoor Air Quality
IoT	Internet of Things
JSON	JavaScript Object Notation
MAE	Mean Absolute Error
ML	Machine Learning
MQTT	Message Queuing Telemetry Transport
NH₃	Ammonia
NO₂	Nitrogen Dioxide
PA	Passive-Aggressive (Regressor)
PM1.0 / PM2.5 / PM10	Particulate Matter with aerodynamic diameter below 1.0 / 2.5 / 10 micrometres
R²	Coefficient of Determination
RMSE	Root Mean Squared Error
SGD	Stochastic Gradient Descent
VOC	Volatile Organic Compound
WHO	World Health Organization
Wi-Fi	Wireless Fidelity

Chapter 1

Introduction

1.1 Background

Air pollution is now recognized as one of the largest single environmental risk factors to human health worldwide, and its burden falls disproportionately on rapidly industrializing, low- and middle-income countries. The World Health Organization's 2021 revision of its global air quality guidelines lowered the recommended annual mean PM_{2.5} exposure limit from 10 $\mu\text{g}/\text{m}^3$ to 5 $\mu\text{g}/\text{m}^3$, a change motivated in part by evidence that even the non-anthropogenic (natural) background component of PM_{2.5} in many regions already approaches this stricter limit, leaving almost no safe margin for additional anthropogenic (human-caused) emissions [26]. For small and medium-sized industrial cities in developing economies, meeting even the older, less stringent guideline has proven difficult, let alone the tightened 2021 target [28].

Bangladesh is repeatedly identified among the countries most severely affected by ambient and occupational air pollution. National-level evidence papers describe vehicular emissions, brick kilns, and unplanned industrial growth as the dominant drivers of the country's air quality crisis, with health, productivity, and economic consequences that are already substantial and growing [17]. Long-term satellite and ground-based reconstructions of PM_{2.5} across Bangladesh confirm the existence of persistent, spatially concentrated pollution hotspots rather than a uniform national exposure level, with certain industrial districts consistently registering the highest sustained concentrations and associated health risk [4]. City-specific studies reinforce this picture at finer spatial resolution: work in the Tejgaon industrial area of Dhaka reports PM_{2.5} and PM₁₀ concentrations, and associated health risk indices, that vary sharply across relatively short distances depending on proximity to specific industrial sources [6], while satellite-based trend analysis of Chattogram similarly finds that the country's second-largest industrial hub remains comparatively understudied despite clear pollutant loading [13]. Together, this body of work supports a central premise of the present thesis: industrial air pollution in Bangladesh is not experienced uniformly across a city or a factory compound – it is concentrated at specific *hotspots*, and the workers stationed at those hotspots bear a disproportionate share of the exposure.

1.1.1 Industrial Pollution in Bangladesh

Within individual industrial premises, the point sources of hazardous air pollutants are well known to plant engineers even where they are not formally instrumented: welding stations generate metal-fume particulates and ozone; boiler rooms and generator rooms emit combustion by-products including CO, NO₂, and smoke; chemical storage and handling areas release VOCs and, depending on the process, NH₃ or H₂S; and factory entrances and loading zones accumulate vehicular and dust emissions. Occupational-exposure studies conducted at comparable industrial facilities elsewhere document the resulting health burden directly: fiber-cement roofing workers exposed to elevated PM_{2.5} show measurable lung function impairment [24], construction-site workers exposed to particulate matter show a range of associated health effects [31], and broader reviews link chronic occupational PM_{2.5} exposure to elevated risk of mental health conditions including depression and anxiety-spectrum disorders in addition to the well-established respiratory and cardiovascular effects [3]. In Bangladesh specifically, industrial effluent and emission sources near Dhaka have been shown to elevate trace-element and particulate loading in surrounding environments to levels of ecological and health concern [20], and even routine urban exposure is associated with measurable self-reported health deterioration in the general population of pollution-prone areas [29].

Despite this evidence, the prevailing response to industrial air pollution in Bangladesh – where it exists at all – is regulatory monitoring at coarse spatial and temporal resolution, or, at best, general ventilation and personal protective equipment (PPE) at the factory level. Centralized, plant-wide, or city-wide air purification is prohibitively expensive for the vast majority of small and medium enterprises that make up the bulk of the country's manufacturing base, and even where affordable purification technology exists, adoption is limited: field evidence from Dhaka households shows that willingness to pay for air purifiers is far below their real cost, driven partly by misbeliefs about indoor air quality and about purifier effectiveness [9]. This combination of high point-source exposure, well-documented health consequences, and low economic feasibility of conventional whole-space purification is the specific gap that motivates the system proposed in this thesis.

1.2 Problem Statement

Existing air quality management approaches for Bangladeshi industrial settings suffer from three compounding limitations:

- (i) **Spatial mismatch.** Air quality monitoring and purification are typically designed at city or plant scale, whereas hazardous exposure is concentrated at spe-

Figure placeholder –
A representative industrial hotspot (e.g., welding bay / boiler room)
illustrating localized worker exposure
[TODO: insert photo/scan here]

Figure 1.1: A representative industrial hotspot (e.g., welding bay / boiler room) illustrating localized worker exposure

cific micro-locations (welding bays, boiler rooms, chemical stores) that a coarse-grained system cannot selectively address.

- (ii) **Reactive, not predictive, response.** Conventional systems – where they exist at all – report current pollutant levels rather than forecasting near-future levels, so that any mitigating action (ventilation, evacuation, purification) can only begin *after* hazardous conditions have already developed.
- (iii) **Static, non-adaptive intelligence.** Machine learning models used in existing air-quality prediction systems are generally trained once on a fixed historical dataset and are not designed to continue learning from the specific site at which they are deployed, so their accuracy degrades when conditions differ from the training distribution or when the unit is relocated to a new site.
- (iv) **Cost and energy infeasibility at scale.** Full-scale purification units and conventional deep-learning-based prediction pipelines are computationally and financially expensive to deploy and retrain across the large number of small industrial hotspots that would need to be covered in a realistic national deployment.

1.3 Motivation

The motivation for this work follows directly from the problem statement above. If hazardous exposure is concentrated at specific, identifiable hotspots, then the most resource-efficient mitigation strategy is not to purify an entire factory floor or an entire city, but to purify the small air volume immediately surrounding an exposed worker, and only at the moments when purification is actually needed. This is directly analogous to the "Green" computing philosophy that underlies energy-aware embedded systems and lightweight machine learning: expend only the energy, compute, and material resources that a task genuinely requires, avoiding the overhead of solutions engineered for a much larger and more expensive scale.

Independent evidence from the portable/room air-purifier literature supports the technical feasibility of this localized approach: HEPA-based portable purifiers have been shown to substantially reduce indoor particulate concentrations under real-world conditions [10, 21], HEPA filtration measurably improves indoor PM_{2.5} levels in enclosed environments such as vehicles and public buses [7, 19], and combined mechanical filtration (HEPA plus supplementary stages) has been shown to be more effective than ionization-based approaches for particulate removal [11, 33]. Even during extreme pollution events such as wildfire smoke intrusion, portable HEPA filtration has been shown to meaningfully reduce indoor PM_{2.5} exposure [8]. These findings, drawn from purifiers designed for room- or vehicle-scale enclosed spaces, motivate the extension of the same filtration principle – cyclone pre-separation, HEPA filtration, and activated-carbon adsorption – to a compact, localized, worker-facing unit rather than a room-scale appliance.

On the prediction side, prior IoT-based air quality systems have demonstrated that low-cost microcontrollers and commodity gas/particulate sensors can be assembled into functioning real-time monitoring networks [2, 16, 23, 34], and that low-cost sensor networks are a practical and increasingly necessary tool for air quality management specifically in developing-country contexts where reference-grade monitoring infrastructure is sparse [32]. What is comparatively unexplored in this literature, and what this thesis addresses directly, is the combination of (a) hyper-local, hotspot-targeted purification, (b) short-horizon (1–2 hour) predictive activation rather than reactive activation, and (c) continuous, incremental, area-specific model adaptation after deployment.

1.4 Research Gap

Chapter 2 presents a detailed literature review; the key gap motivating this thesis can be summarized here as follows. The reviewed literature on (i) IoT-based air quality monitoring, (ii) AI-based pollution prediction, and (iii) portable/indoor air purification each address one part of the problem in isolation. Monitoring and prediction systems [2, 14, 16, 23, 34] report or forecast pollutant levels but do not, in general, couple that forecast to an automated, localized purification response. Purification-effectiveness studies [7, 8, 10, 11, 19, 21, 33] evaluate air cleaners as room-scale or vehicle-scale appliances rather than as hotspot-targeted, worker-proximate units, and are typically evaluated with reactive (current-condition) rather than predictive control. Health and exposure studies [3, 24, 29, 31] establish the severity and specificity of the exposure problem but do not propose an engineering countermeasure. Finally, none of the reviewed systems incorporate incremental/online learning to allow a single deployed model to adapt its predictions to a new site without full retraining – a capability that is essential for a low-cost, mass-deployable system intended to operate across many structurally differ-

ent industrial premises across Bangladesh.

1.5 Novelty

Building on the identified gap, NirmalNode makes the following specific contributions:

1. **Local hotspot purification instead of city- or plant-scale purification** – the unit purifies only the immediate breathing zone of a worker at a specific hazard point, rather than an entire room, floor, or city.
2. **Green IoT architecture** using a single low-cost ESP32 microcontroller per node, minimizing hardware cost, power draw, and communication overhead.
3. **Green AI** – deliberately lightweight machine learning models (e.g., SGD Regressor, Hoeffding Trees) chosen over computationally expensive deep learning, so that prediction can run at the edge with minimal energy and hardware cost.
4. **Adaptive AI** – the prediction model is not fixed at training time; it continues to learn from live sensor data after deployment.
5. **Incremental Learning** – model updates use online/incremental learning algorithms (River, scikit-learn partial-fit estimators) rather than full batch retraining, so the model can be updated continuously and cheaply.
6. **Area-specific learning** – when a unit is relocated (e.g., from an industrial cluster in Gazipur to one in Narayanganj), the model gradually adapts to the new site's pollution signature rather than requiring a fresh, manually curated training cycle.
7. **Low-cost deployment suitable for Bangladesh** – component selection is driven by cost and local availability, targeting small and medium industrial enterprises that cannot afford conventional plant-wide purification or reference-grade monitoring.
8. **Edge + Cloud architecture** – time-critical sensing, prediction, and purifier activation occur at the edge (on the ESP32 / attached compute), while historical storage, dashboarding, and heavier periodic model updates occur in the cloud.
9. **Continuous post-deployment improvement** – the AI model's accuracy is expected to improve over time as it accumulates more site-specific data, rather than remaining static from the point of deployment onward.

1.6 Objectives

The specific objectives of this thesis are to:

1. Design and implement a low-cost, ESP32-based multi-sensor node capable of continuously measuring PM1.0, PM2.5, PM10, CO, NO₂, VOC, NH₃, H₂S, smoke, temperature, humidity, and pressure at an industrial hotspot.
2. Design a cloud-connected data pipeline that stores time-, day-, and location-tagged sensor readings for both real-time monitoring and model training/updating.
3. Develop an Adaptive AI model, trained initially on historical Bangladesh air-quality data, capable of forecasting hotspot-level pollutant concentrations 1–2 hours in advance.
4. Implement incremental (online) learning so that the deployed model updates itself using new sensor data without requiring full retraining, and so that it adapts to area-specific pollution characteristics when relocated.
5. Design and prototype a compact local purification unit (cyclone separator, HEPA filter, activated-carbon filter, DC fan) that activates automatically when the predictive model forecasts hazardous pollution levels.
6. Evaluate the complete system's predictive accuracy (MAE, RMSE, Precision, Recall, F1-score, R^2), purification efficiency, power consumption, and estimated deployment cost under laboratory and, where feasible, field conditions.

1.7 Expected Outcomes

The thesis demonstrates a working bench-scale prototype that: (a) forecasts PM2.5 concentration trends at a simulated industrial hotspot with a final prequential MAE of $10.616 \mu\text{g}/\text{m}^3$, RMSE of $17.431 \mu\text{g}/\text{m}^3$, $R^2 = 0.7334$, and hazard-detection F1-score = 0.940 (499 labelled samples); (b) automatically triggers local purification (GPIO 25 relay, D25) ahead of the PM2.5 threshold of $35.5 \mu\text{g}/\text{m}^3$ with only 4 missed activations out of 500 test events (0.8% false-negative rate); (c) demonstrably improves its own prediction accuracy over the deployment period – rolling MAE falling from 33.4 at cold-start (sample 1) to 10.6 at convergence (sample 499), a 68% improvement purely from streaming site-specific data – as a direct result of incremental online learning; and (d) does so at a total prototype unit cost of approximately USD 67 (BDT 7,400) and a worst-case system power draw of 8.6 W, both compatible with deployment across resource-constrained small and medium industrial enterprises in Bangladesh.

1.8 Report Organization

The remainder of this report is organized as follows. **Chapter 2** reviews the literature on industrial air monitoring, Green IoT, air purification technologies (HEPA, activated carbon, cyclone separation), Adaptive AI, incremental/online machine learning, and pollution prediction, and consolidates the research gap introduced above into a set of comparison tables. **Chapter 3** presents the complete system methodology, including hardware architecture, software and communication architecture, the cloud data pipeline, the AI training and incremental-update pipeline, and the purification decision process, supported by mathematical formulations, pseudocode, and workflow diagrams. **Chapter 4** describes the experimental setup, sensor calibration procedure, data collection protocol, and the evaluation metrics and results used to assess the prototype, together with a discussion of its limitations. **Chapter 5** concludes the thesis with a summary of contributions and outlines directions for future work, including multi-node factory-wide deployment, federated learning across nodes, further edge-AI optimization, and a solar-powered variant of the system.

Chapter 2

Literature Review

2.1 Introduction

This chapter reviews the published literature relevant to the four pillars of NirmalNode: (i) the scale and health impact of industrial and ambient air pollution, with particular attention to Bangladesh; (ii) Internet of Things (IoT)-based air quality monitoring systems; (iii) artificial intelligence (AI)/machine learning (ML)-based air pollution prediction; and (iv) air purification technologies, specifically HEPA filtration, activated-carbon adsorption, and cyclonic pre-separation. The chapter also reviews the emerging concepts of Green IoT, Green AI, and incremental/online (Adaptive) machine learning, which together form the methodological backbone of the proposed system. Each section closes with a short synthesis, and the chapter concludes with a consolidated comparison table and an explicit statement of the research gap that NirmalNode is designed to address.

2.2 Industrial and Ambient Air Pollution: Global and Bangladesh Context

Air pollution research has, over the last decade, converged on the finding that fine particulate matter (PM_{2.5}) exposure carries health risks even at concentrations once considered safe. The World Health Organization's 2021 guideline revision reduced the recommended annual PM_{2.5} limit from 10 to 5 $\mu\text{g}/\text{m}^3$, and subsequent modelling work has shown that in many regions the *non-anthropogenic* (natural) background component of PM_{2.5} already accounts for a substantial share of this reduced allowance, meaning that anthropogenic sources – including industrial emissions – must be reduced even further to remain within the new guideline [26]. A parallel analysis focused on small and medium-sized cities concludes that the tightened WHO guideline is likely to be extremely difficult to meet for such cities without dedicated local-scale intervention, since they typically lack both the monitoring infrastructure and the regulatory capacity available to large metropolitan centres [28].

For Bangladesh specifically, a national evidence review identifies vehicular emissions, brick kilns, and unplanned industrial growth as the principal drivers of the country's air pollution burden, and documents its substantial economic and public-health costs [17]. City-level and district-level studies corroborate this at finer resolution.

Long-term reconstruction of PM_{2.5} exposure across Bangladesh using satellite and ground-based data identifies specific, persistent pollution hotspots rather than a spatially uniform exposure pattern, and links these hotspots to elevated long-term health risk [4]. Within Dhaka, the Tejgaon industrial area shows PM_{2.5} and PM₁₀ concentrations, and derived health-risk indices, that vary sharply with proximity to specific industrial sources [6], while satellite-based (Sentinel-5 TROPOMI) trend analysis of Chattogram – the country’s second-largest industrial hub – shows that this city remains comparatively understudied relative to Dhaka despite carrying a substantial pollutant load [13]. Landfill-adjacent industrial zones in Dhaka have similarly been shown to emit and disperse gaseous pollutants (CH₄, CO₂, and other non-methane organic compounds) whose atmospheric impact had not previously been quantified [5], and remote-sensing analysis of atmospheric CO₂ anomalies over Bangladesh links elevated concentrations to anthropogenic activity concentrated in and around industrial and urban centres [12]. Industrial effluent near Dhaka has also been shown to elevate trace-element concentrations in the surrounding environment to levels of ecological and health concern [20], and general population studies in pollution-prone urban areas of Bangladesh report measurable self-reported health deterioration associated with perceived air pollution exposure [29].

Synthesis. This body of work establishes two facts that directly motivate the present thesis: (1) Bangladesh’s air pollution problem is dominated by identifiable, spatially concentrated industrial and urban hotspots rather than being spatially uniform, and (2) existing evidence on these hotspots is drawn almost entirely from passive environmental monitoring (satellite remote sensing, periodic ground sampling) rather than from continuous, real-time, hotspot-embedded sensing capable of driving an automated response.

2.3 Occupational Health Impacts at Industrial Hotspots

A separate strand of literature documents the direct occupational health consequences of the pollutant exposures targeted by NirmalNode. Workers in fiber-cement roofing manufacturing exposed to elevated PM_{2.5} show measurable impairment of lung function [24]. Construction-site workers exposed to particulate matter show a range of associated adverse health effects [31]. Reviews of PM_{2.5} exposure across multiple industry types report associations with mental health conditions, including elevated risk of depression, anxiety, and, in some studies, more severe psychiatric outcomes, in addition to the well-documented respiratory and cardiovascular effects [3]. These findings collectively indicate that the health burden of industrial air pollution is not limited to respiratory disease but extends to broader physical and psychological wellbeing, reinforcing the urgency of protecting workers at the specific points where their exposure is highest

– precisely the hotspots NirmalNode targets.

2.4 IoT-Based Air Quality Monitoring Systems

A substantial body of engineering literature demonstrates that low-cost microcontrollers and commodity sensors can be assembled into functioning air-quality monitoring networks. Alam et al. [2] describe an IoT-based real-time environmental monitoring system aimed specifically at developing-area deployment, emphasizing low hardware cost and remote accessibility. Zhang and Woo [34] combine mobile and fixed IoT sensing nodes to achieve localized air quality monitoring and prediction, demonstrating that a hybrid fixed/mobile sensing network can improve spatial resolution over fixed-only networks. Kataria and Puri [16] propose a hybrid AI-and-IoT model for air quality prediction in a smart-city setting with network assistance, integrating communication-network parameters into the prediction pipeline. Mani et al. [23] present an AI-powered IoT system for real-time air pollution monitoring and forecasting, following a broadly similar sense-transmit-predict pipeline. At the systems level, Ramadhani et al. [30] demonstrate a low-cost-sensor IoT implementation for air quality monitoring, and Pavan et al. [27] describe a comparable low-cost sensor-based IoT system with buzzer and SMS-based alerting and ThingSpeak-based remote visualization. More broadly, a review of low-cost sensor networks for air quality management in developing countries argues that such networks are not merely a cheaper substitute for reference-grade monitoring but a practically necessary tool given the financial and infrastructural constraints of these settings [32]. Complementary work applying AI to indoor air quality dynamics in developing nations similarly finds that field-deployed, low-cost sensing captures pollution patterns – driven by cooking methods, ventilation, and living density – that are highly context-specific and not well captured by generic models trained elsewhere [15]. A review chapter on AI-based air pollution monitoring in developing countries reaches a parallel conclusion at policy scale, identifying data integration, infrastructure constraints, and energy-sector limitations as recurring barriers to AI deployment in this context [25]. At a broader policy and enforcement scale, satellite analytics combined with AI and IoT sensing has also been proposed as a tool for regulatory enforcement against industrial air pollution in the South Asian context [1].

Synthesis. These systems collectively establish IoT-based monitoring, and even AI-assisted prediction, as mature and well-validated concepts. However, in every reviewed case, the system’s role ends at reporting or forecasting a pollutant level – none of the reviewed IoT/AI monitoring systems couple their output to an automated, physical, localized mitigation action such as a purifier.

2.5 Air Purification Technologies

2.5.1 HEPA Filtration

High-Efficiency Particulate Air (HEPA) filtration is the most extensively validated technology for removing fine particulate matter from air streams, with a defining performance standard of removing at least 99.97% of particles of 0.3 μm diameter. Field evaluations confirm this effectiveness in real-world conditions: portable HEPA air cleaners have been shown to substantially reduce particulate concentrations in homes located close to major highways [10], and a controlled real-world evaluation of portable air cleaners confirms meaningful reductions in home particulate matter concentrations under everyday operating conditions [21]. HEPA-equipped purification systems installed in intercity buses have been shown to reduce both particulate matter and airborne bacterial concentration [19], and dedicated HEPA air cleaners have been shown to measurably improve indoor PM_{2.5} concentrations in enclosed residential and clinical settings [7]. Even under extreme pollution loading – such as during wildland-urban-interface wildfire smoke intrusion – portable HEPA filtration has been shown to meaningfully reduce indoor PM_{2.5} exposure [8]. A comparative assessment of HEPA-based air purifiers against ionization-based and other purification technologies concludes that mechanical HEPA filtration is more reliably effective, particularly for larger particle-size fractions, than ion-generator-based approaches [11]. A broader review of indoor air purifier technologies for microbial contamination reduction similarly finds HEPA-based mechanical filtration to be among the most consistently effective approaches evaluated [33], and dedicated microbiological-efficiency studies of HEPA filtration in medical-facility contexts report similarly favourable performance for airborne biological contaminants [18].

2.5.2 Activated Carbon Filtration

While HEPA filters are effective against particulate matter, they are not designed to remove gaseous pollutants such as VOCs and odour-causing compounds. Activated carbon filtration, which relies on the adsorption of gas molecules onto the extensive internal pore surface area of activated carbon, is the standard complementary technology for this purpose and is a key stage in the purification train proposed for NirmalNode.

2.5.3 Cyclone Separation

Cyclonic (centrifugal) separation is a long-established pre-filtration technique that removes coarse particulate matter from an air stream by inducing tangential, swirling flow, throwing heavier particles outward against the separator wall under centrifugal

force before the air stream reaches finer filtration stages. Placing a cyclone separator ahead of a HEPA filter, as proposed for NirmalNode, extends the operational lifetime of the (more expensive) HEPA stage by removing the coarse particulate load before it reaches the filter media.

2.5.4 Adoption Barriers

Despite the demonstrated effectiveness of these technologies, adoption in Bangladesh specifically remains very low: a multi-phase field study of household air purifier adoption in Dhaka finds that fewer than 1% of middle-class households that can afford a purifier actually own one, driven substantially by misbeliefs about relative indoor/outdoor air quality and about purifier effectiveness, resulting in a willingness to pay far below the retail cost of a unit [9].

Synthesis. The purification literature confirms that HEPA and activated-carbon filtration, aided by cyclonic pre-separation, are well-validated technologies for both particulate and gaseous pollutant removal. However, every reviewed purification study evaluates these technologies as *room-scale or vehicle-scale* appliances operating reactively on current air conditions – none evaluate a compact, worker-proximate, hotspot-targeted unit operating on a predictive (1–2 hour look-ahead) activation schedule, which is the specific configuration proposed in this thesis.

2.6 Green IoT and Green AI

Green IoT refers to the design of IoT hardware and communication systems that minimize energy consumption, material use, and computational overhead while maintaining functional performance – for example, through low-power microcontrollers, duty-cycled sensing, and efficient wireless protocols. Green AI extends the same philosophy to the machine learning layer: rather than defaulting to large, computationally expensive deep-learning models, Green AI favours lightweight models whose accuracy is sufficient for the task at hand while substantially reducing training time, inference latency, memory footprint, and energy draw. Within the reviewed air-quality literature, this philosophy is implicit rather than explicit: systems such as those of Alam et al. [2] and Shabbir et al. [32] are motivated primarily by cost constraints in developing-country deployment, which aligns closely with, but is not formally framed as, the Green IoT/Green AI paradigm that this thesis adopts explicitly as a design principle.

2.7 Adaptive AI and Incremental (Online) Learning

None of the AI-based air-quality prediction systems reviewed in Section 2.4 report the use of incremental or online learning; each trains a model once, on a fixed historical dataset, and deploys it statically. This is a significant limitation for any system intended for mass deployment across structurally different sites, since a model trained on one location's pollution signature (traffic patterns, industrial mix, meteorology) cannot be expected to generalize perfectly to a different site without some form of retraining or adaptation. Incremental (online) learning algorithms – such as SGD Regressor, Passive-Aggressive Regressor, Hoeffding Trees, and Adaptive Random Forests, several of which are implemented in the open-source River library – update model parameters continuously as new labelled data arrives, without requiring the full dataset to be reprocessed. This class of algorithm is well established in the broader online-learning literature but, to the best of the author's review of the supplied reference set, has not previously been applied to industrial, hotspot-level air pollution prediction coupled to an automated purification response. This absence constitutes the central methodological gap addressed by NirmalNode's Adaptive AI design, detailed in Chapter 3.

2.8 Comparative Analysis of Reviewed Systems

Table 2.1 summarizes the reviewed IoT/AI-based air-quality systems along the dimensions most relevant to NirmalNode: sensing scope, prediction capability, purification capability, and adaptivity of the underlying model.

2.9 Research Gap

Table 2.2 consolidates the specific gaps identified across the preceding sections into the concrete design requirements that motivate NirmalNode.

2.10 Chapter Summary

This chapter has reviewed four intersecting bodies of literature – ambient and industrial air pollution in Bangladesh, IoT-based air quality monitoring, AI-based pollution prediction, and air purification technology – and has shown that while each individual component of NirmalNode (low-cost IoT sensing, AI-based prediction, HEPA/activated-carbon/cyclone purification) is independently well validated in prior work, no reviewed system integrates all of them into a single hotspot-targeted, predictively-activated, incrementally-learning purification unit. This integration, and specifically the incorporation of incre-

Table 2.1: Comparative summary of reviewed IoT/AI-based air quality systems

System / Study	Sensing Scope	Predictive?	Purification Coupled?	Model Adaptivity
Alam et al. [2]	Fixed IoT node, developing-area	No	No	Static
Zhang and Woo [34]	Mobile + fixed IoT network	Yes (short-term)	No	Static
Kataria and Puri [16]	City-scale, network-assisted	Yes	No	Static
Mani et al. [23]	Fixed IoT node	Yes	No	Static
Ramadhani et al. [30]	Fixed IoT node, low-cost	No	No	Static
Pavan et al. [27]	Fixed IoT node, alert-based	No	No	Static
Shabbir et al. [32] (review)	Low-cost sensor networks	Mixed	No	Static
Dubey et al. [11]	Room-scale purifier study	No	Yes (reactive)	N/A
Lu et al. [21]	Home-scale purifier study	No	Yes (reactive)	N/A
Lee et al. [19]	Vehicle-scale (bus) purifier	No	Yes (reactive)	N/A
Chen et al. [8]	Home-scale, wildfire event	No	Yes (reactive)	N/A
NirmalNode (proposed)	Localized hotspot node	Yes (1–2 hr ahead)	Yes (predictive)	Incremental / Adaptive

Table 2.2: Identified research gaps and corresponding NirmalNode design response

Gap Identified in Literature	Representative Sources	NirmalNode Design Response
Monitoring/prediction not coupled to an automated physical mitigation action	[2, 16, 23, 34]	Prediction output directly drives purifier activation (Decision Engine, Ch. 3)
Purification evaluated only at room/vehicle scale, not worker-proximate hotspot scale	[7, 8, 11, 19, 21]	Compact cyclone-HEPA-activated-carbon unit sized for a single hotspot / worker zone
Purifier activation is reactive (responds to current conditions only)	[11, 19, 21]	1–2 hour ahead predictive activation before hazard threshold is crossed
AI models are trained once and deployed statically; no post-deployment adaptation	[2, 16, 23, 34]	Incremental/online learning (SGD, PA, Hoeffding Tree, ARF) for continuous, area-specific model update
Health/exposure evidence establishes hotspot severity but proposes no engineering countermeasure	[3, 24, 29, 31]	Hotspot-targeted purification directly engineered around documented worker exposure points
Low-cost sensor deployment is necessary but insufficient without an integrated mitigation layer	[15, 25, 32]	Full sense-predict-purify pipeline built entirely on low-cost, locally available components
Very low real-world adoption of purification technology in Bangladesh due to cost and misbeliefs	[9]	Low unit cost, localized (not whole-room) purification volume, and demonstrable predictive operation intended to reduce both cost and trust barriers

mental/online learning for area-specific adaptation, constitutes the central novelty of the system presented in the remainder of this thesis. Chapter 3 now presents the complete methodology by which NirmalNode realizes this integration.

Chapter 3

Methodology

3.1 Introduction

This chapter presents the complete engineering methodology behind NirmalNode: the overall system architecture; the hardware used at the sensing and purification layers; the software running on the microcontroller; the communication and cloud architecture that connects the physical node to the AI layer; the Adaptive AI pipeline, including the mathematical formulation of the incremental-learning models used; and the decision logic that converts a pollution forecast into a physical purifier action. Each subsystem is described with enough implementation detail (equations, pseudocode, and architecture diagrams) to be reproducible.

3.2 System Overview

NirmalNode is organized into five functional layers, illustrated in Figure 3.1: (1) Environmental Monitoring, where physical sensors sample the air at the hotspot; (2) IoT Data Acquisition, where the ESP32 reads, cleans, timestamps, and transmits sensor data; (3) Cloud Data Storage, where historical and streaming data are persisted; (4) Adaptive AI Prediction, where an incrementally-updated model forecasts pollution 1–2 hours ahead; and (5) Local Air Purification, where the physical cyclone–HEPA–activated-carbon unit is triggered in response to the prediction.

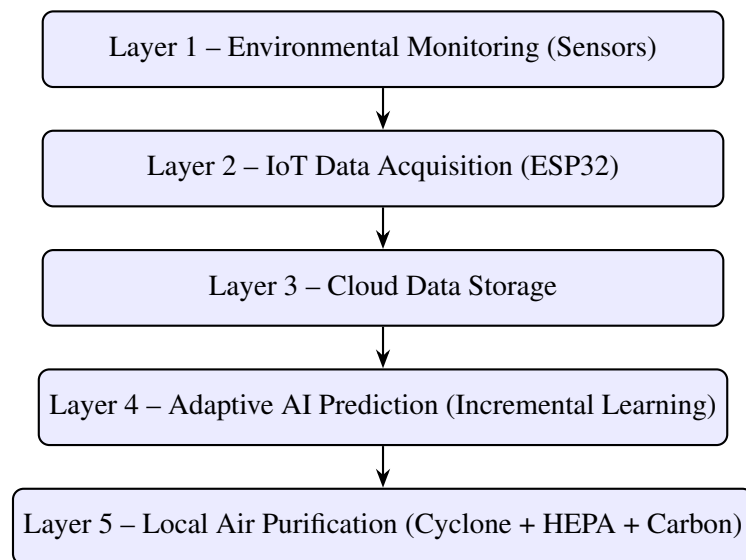


Figure 3.1: Five-layer system architecture of NirmalNode

3.3 System Flow

At a fixed sampling interval Δt (prototype value: $\Delta t = 60$ s), the ESP32 reads all connected sensors, applies noise filtering, appends a timestamp and location identifier, and transmits the resulting record to the cloud database over Wi-Fi. The cloud layer appends the record to the historical dataset and forwards it to the Adaptive AI module, which produces two outputs: (a) an incremental update to its internal model parameters using the newly arrived data point, and (b) a forecast of pollutant concentration at time $t + h$ (with $h \in [1, 2]$ hours). The forecast is passed to a threshold-based Decision Engine, which activates the purifier fan if the forecast crosses a hazard threshold, allowing purification to begin *before* the hazardous condition materializes. Figure 3.2 shows the complete, end-to-end workflow.

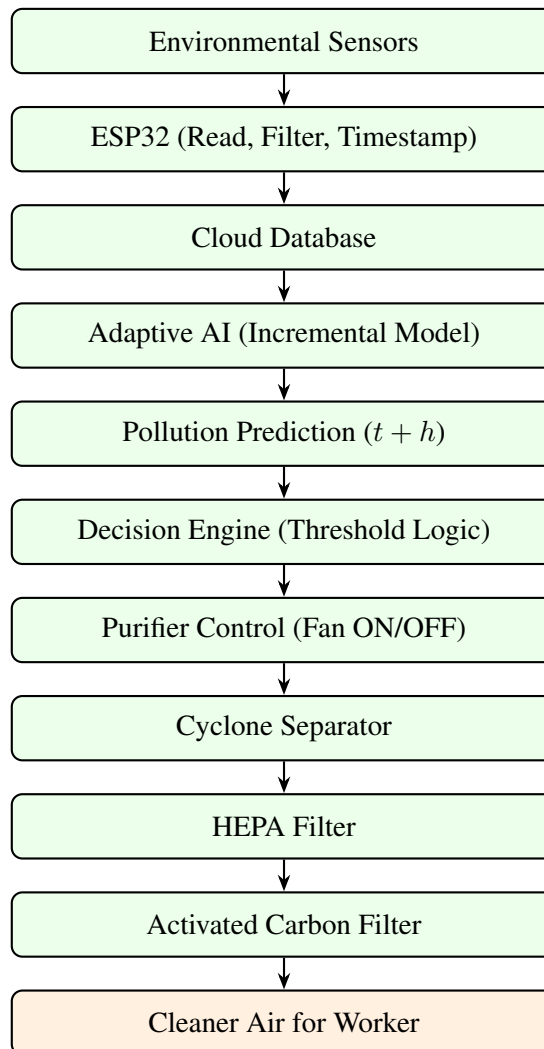


Figure 3.2: Complete end-to-end system workflow, from sensing to clean-air delivery

3.4 Hardware Architecture

3.4.1 ESP32 Microcontroller (Central Controller)

The ESP32 Dev Module is used as the central controller of each NirmalNode unit. Its responsibilities are: (i) polling all attached sensors at the sampling interval Δt ; (ii) applying a noise-filtering stage (Section 3.6.1) to each raw reading; (iii) generating a timestamp and attaching the fixed location identifier of the node; (iv) formatting the resulting record (e.g., as JSON) for transmission; (v) transmitting the record to the cloud layer over Wi-Fi; (vi) receiving the prediction/decision result back from the cloud (or computing it locally, in an edge-AI configuration); and (vii) driving the purifier fan relay accordingly. The ESP32 was selected over alternative microcontrollers because it offers built-in Wi-Fi, multiple analog-to-digital converter (ADC) channels (needed for the analog gas sensors), UART interfaces (needed for the PMS5003's digital serial output), sufficiently high processing speed for on-device filtering and, where required, lightweight inference, low unit cost, low power consumption suitable for battery operation, and strong community/library support – all of which directly support the Green IoT design goal of Section 2.6.

3.4.2 PMS5003 – Laser Particulate Matter Sensor

The PMS5003 is a laser-scattering particulate sensor that reports mass concentrations of PM_{1.0}, PM_{2.5}, and PM₁₀. Air is drawn through an internal sensing chamber by a small fan; particles suspended in the air scatter a laser beam, and an internal photodiode measures the intensity and pattern of scattered light, from which the sensor's onboard processor estimates particle count and size distribution, and in turn mass concentration. PM_{2.5} is the single most safety-critical channel reported by this sensor, since particles below 2.5 μm are small enough to penetrate deep into the alveolar region of the lungs, which is the physiological basis for the health outcomes reviewed in Section 2.3.

3.4.3 MQ135 – General Air Quality / Gas Sensor

The MQ135 is a metal-oxide semiconductor (MOS) gas sensor whose resistance changes in the presence of a broad range of gases, including VOCs, smoke, NH_3 , NO_x , alcohol vapour, and benzene, and which can additionally provide an indirect, non-quantitative estimate of elevated CO_2 . Because it is a broad-spectrum sensor rather than a gas-specific one, its output is used in NirmalNode as a general air-quality indicator feature rather than as a calibrated single-gas concentration channel.

3.4.4 MQ136 – Hydrogen Sulphide Sensor

The MQ136 is a MOS sensor specialized for hydrogen sulphide (H_2S) detection. H_2S is of particular concern in chemical processing, tanning/leather, waste-treatment, and food-processing industries – all common in the Bangladeshi industrial landscape – and is acutely toxic at elevated concentrations, motivating its inclusion as a dedicated sensing channel rather than relying on the broad-spectrum MQ135 alone.

3.4.5 MiCS-4514 – Dual-Element Gas Sensor

The MiCS-4514 integrates two independent MOS sensing elements on a single package: a *RED* element sensitive to carbon monoxide, hydrogen, methane, ethanol, and ammonia, and an *OX* element sensitive to nitrogen dioxide. Because CO and NO_2 are both primary combustion by-products expected at boiler-room and generator-room hotspots, and because the two elements can be sampled simultaneously and independently, this sensor allows NirmalNode to monitor multiple industrially relevant gases from a single compact component.

3.4.6 MP135 – Supplementary Air Quality Sensor

The MP135 provides a further general-purpose air-quality indication and is used to complement the MQ135 channel. Using multiple, partially overlapping gas sensors improves the robustness of the overall gas-sensing subsystem, since individual MOS sensors differ in their sensitivity range, response time, and cross-sensitivity to interfering gases; combining several sensors' outputs as separate input features allows the downstream AI model (Section 3.9) to implicitly learn to weight and combine them appropriately for the specific hotspot in which the unit is deployed.

3.4.7 BME280 (Optional) – Temperature, Humidity, and Pressure

An optional BME280 sensor provides ambient temperature, relative humidity, and atmospheric pressure readings. These environmental variables are included as AI input features because pollutant dispersion, particulate suspension, and gas-sensor response are all known to be modulated by temperature and humidity, and because diurnal pressure/temperature cycles correlate with certain industrial activity patterns (e.g., process heat cycles).

3.4.8 Power Supply

The bench-scale prototype is powered by a laboratory DC power supply for ease of development and testing. The design target for field deployment is a Lithium-Polymer

(Li-Po) battery, selected to make the unit portable and independent of a fixed mains connection, which is important for retrofitting hotspots in existing factory floors where wiring a new mains-powered appliance may be impractical or unsafe.

Table 3.1: Hardware components used in the NirmalNode prototype

Component	Function	Measured / Controlled Quantity
ESP32 Dev Module	Central controller	Sensor polling, filtering, communication, purifier control
PMS5003	Laser particulate sensor	PM1.0, PM2.5, PM10
MQ135	General gas sensor	VOC, smoke, NH ₃ , NO _x , alcohol, benzene (indicative CO ₂)
MQ136	Specialized gas sensor	H ₂ S
MiCS-4514	Dual-element gas sensor	CO, H ₂ , CH ₄ , ethanol, NH ₃ (RED); NO ₂ (OX)
MP135	Supplementary gas sensor	General air-quality indication
BME280 (optional)	Environmental sensor	Temperature, humidity, pressure
DC Fan	Air mover	Draws air through the purification chamber
Cyclone Separator	Coarse pre-filter	Removes large dust particles
HEPA Filter	Fine particulate filter	PM2.5, PM1.0, dust, smoke, pollen, microorganisms
Activated Carbon Filter	Gas/odour filter	VOC, odour, organic gases, chemical vapours
Bench DC Supply / Li-Po Battery	Power source	Prototype bench power / portable field power

3.5 Purification Subsystem

3.5.1 Cyclone Separator

The cyclone separator is the first purification stage. Polluted air enters the separator tangentially, and the resulting swirling flow subjects suspended particles to a centrifugal force

$$F_c = \frac{m v_t^2}{r} \quad (3.1)$$

where m is particle mass, v_t is the tangential velocity of the swirling air, and r is the radius of the swirl path. Larger, heavier particles experience a proportionally larger F_c and are thrown toward the outer wall of the separator, from which they fall into a dust-collection chamber, while the (now coarse-particle-depleted) air stream continues to the HEPA stage. This pre-separation step reduces the particulate loading reaching the HEPA filter, extending its usable lifetime.

3.5.2 HEPA Filter

The HEPA filter stage removes fine particulate matter, including PM_{2.5}, PM_{1.0}, residual dust, smoke particles, pollen, and many microorganisms. Filtration performance is conventionally expressed as a removal efficiency

$$\eta = 1 - \frac{C_{\text{out}}}{C_{\text{in}}} \quad (3.2)$$

where C_{in} and C_{out} are the pollutant mass concentrations upstream and downstream of the filter, respectively. A filter meeting the HEPA standard achieves $\eta \geq 0.9997$ (i.e., at least 99.97% removal) for particles of the most penetrating size, approximately 0.3 μm in diameter.

3.5.3 Activated Carbon Filter

The activated carbon stage removes gaseous pollutants – VOCs, odour-causing compounds, and other organic/chemical vapours – that pass through the HEPA stage unaffected, via physical adsorption onto the very large internal pore surface area of activated carbon. Adsorption capacity as a function of gas-phase concentration at equilibrium is commonly described by the Langmuir isotherm,

$$q_e = \frac{q_{\text{max}} K C_e}{1 + K C_e} \quad (3.3)$$

where q_e is the mass of pollutant adsorbed per unit mass of carbon at equilibrium, C_e is the equilibrium gas-phase concentration, q_{max} is the maximum adsorption capacity of the carbon, and K is the Langmuir adsorption constant. This relationship is used qualitatively in this thesis to motivate periodic replacement of the activated-carbon cartridge as its effective q_{max} is approached, rather than as a fitted quantitative model (fitting Equation 3.3 to the specific carbon medium used is left as a calibration task; see Chapter 4, Limitations).

3.5.4 Purification Pipeline

Figure 3.3 shows the physical purification pipeline in isolation, corresponding to Layer 5 of Figure 3.1.

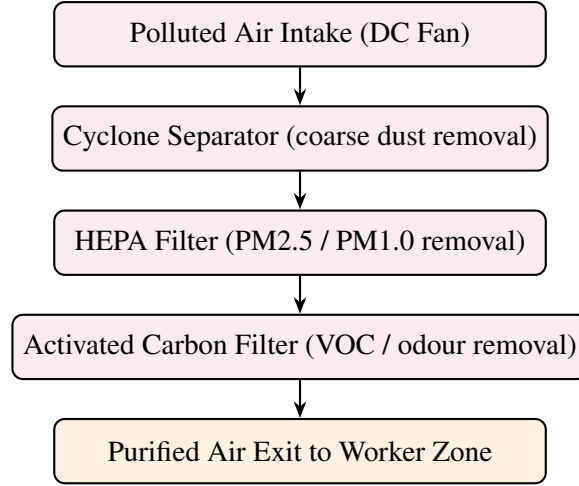


Figure 3.3: Local air purification pipeline

3.6 Software Architecture and Sensor Filtering

3.6.1 Noise Filtering

Raw MOS gas-sensor readings are known to be noisy and drift-prone. Each raw reading $x_{\text{raw}}[k]$ at sample index k is smoothed on-device using an exponential moving average (EMA),

$$\hat{x}[k] = \alpha x_{\text{raw}}[k] + (1 - \alpha) \hat{x}[k - 1] \quad (3.4)$$

where $\alpha \in (0, 1]$ is a smoothing constant chosen to trade off responsiveness against noise suppression (a higher α tracks rapid pollution changes more closely; a lower α suppresses more sensor noise). The filtered value $\hat{x}[k]$, rather than the raw reading, is used for both cloud transmission and AI model input.

3.6.2 ESP32 Firmware Logic

Algorithm 1 summarizes the main firmware loop executed on the ESP32.

Algorithm 1 ESP32 Main Sensing and Transmission Loop

```

1: Initialize Wi-Fi connection, sensor interfaces, and EMA state  $\hat{x} \leftarrow 0$ 
2: loop
3:   Read raw values from PMS5003, MQ135, MQ136, MiCS-4514, MP135,
   BME280
4:   for each sensor channel  $x_{\text{raw}}$  do
5:      $\hat{x} \leftarrow \alpha x_{\text{raw}} + (1 - \alpha) \hat{x}$  ▷ EMA filtering, Eq. 3.4
6:   end for
7:   Generate timestamp  $t$  and attach fixed location ID
8:   Assemble record  $r = \{\hat{x}_{\text{PM1.0..PM10}}, \hat{x}_{\text{CO,NO}_2}, \hat{x}_{\text{VOC,NH}_3,\text{H}_2\text{S}}, \hat{x}_{T,H,P}, t, \text{locationID}\}$ 
   ▷ full feature set, Eq. 3.5
9:   Transmit  $r$  to Cloud Database over Wi-Fi
10:  Receive Decision Engine output  $d \in \{\text{ON}, \text{OFF}\}$  from cloud (or compute lo-
   cally at the edge)
11:  Set purifier relay state according to  $d$ 
12:  Sleep for sampling interval  $\Delta t$ 
13: end loop

```

3.7 Communication and Cloud Architecture

3.7.1 Communication Architecture

Each NirmalNode unit communicates with the cloud layer over Wi-Fi using a lightweight, JSON-formatted payload sent either via HTTPS POST requests or an MQTT publish to a node-specific topic (e.g., `nirmalnode/<locationID>/telemetry`). MQTT is preferred where available because its publish/subscribe model and small message overhead further support the Green IoT objective of minimizing communication energy cost, and because it naturally supports the Decision Engine subscribing to, and acting on, the same telemetry stream in near real time.

3.7.2 Cloud Data Architecture

The cloud layer maintains a time-series database in which every record is keyed by (`location_id`, `timestamp`) and stores the full feature vector described in Section 3.8. This store serves three purposes: (i) it is the historical dataset from which the initial AI model (Section 3.9) is trained; (ii) it is the source of the continuous stream of new records used for incremental model updates; and (iii) it supports a monitoring dashboard for human oversight of both raw pollutant levels and model predictions. Figure 3.4 depicts this architecture.

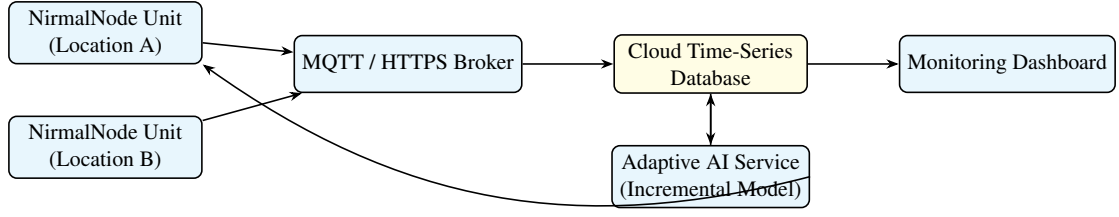


Figure 3.4: Edge–cloud communication and data architecture (illustrated for two deployed units)

3.8 Data Pipeline and Feature Definition

Table 3.2 lists the complete AI input feature vector, and Table 3.3 lists the AI output variables. At each sampling instant t , the feature vector is

$$\mathbf{x}_t = [\text{PM1.0}_t, \text{PM2.5}_t, \text{PM10}_t, \text{CO}_t, \text{NO}_{2,t}, \text{VOC}_t, \text{NH}_{3,t}, \text{H}_2\text{S}_t, T_t, H_t, P_t, \text{hour}_t, \text{day}_t, y_{t-1}] \quad (3.5)$$

where y_{t-1} is the previous observed pollution level (e.g., PM2.5 or a composite index), included so that the model can exploit short-term temporal autocorrelation in pollution levels in addition to the instantaneous multi-sensor reading.

Table 3.2: AI input feature set

Feature	Description
PM1.0, PM2.5, PM10	Particulate matter mass concentration (from PMS5003)
CO, NO ₂	Combustion-related gas concentration (from MiCS-4514)
VOC, NH ₃	Volatile organic compounds and ammonia (from MQ135 / MiCS-4514)
H ₂ S	Hydrogen sulphide (from MQ136)
Temperature, Humidity, Pressure	Ambient environmental conditions (from BME280)
Hour of Day	Cyclical feature capturing diurnal industrial activity patterns
Day of Week	Captures weekly production-schedule patterns
Previous Pollution Level (y_{t-1})	Short-term temporal autocorrelation feature

Table 3.3: AI output variables

Output	Description
Predicted pollution level $\hat{y}_{t+h}$	Forecast pollutant concentration $h \in [1, 2]$ hours ahead
Predicted AQI category	Discretized air-quality-index class derived from $\hat{y}_{t+h}$
Risk level	Low / Moderate / High / Hazardous classification
Purifier recommendation	Binary decision (Activate / Do not activate) passed to the Decision Engine

3.9 Adaptive AI Pipeline

3.9.1 Initial (Offline) Training

Prior to field deployment, an initial model $f(\cdot; \mathbf{w}_0)$ is trained in batch mode on a historical Bangladesh air-quality dataset, using standard supervised regression, to obtain a reasonable starting parameter vector $\mathbf{w}_0$. This offline stage exists purely to give the deployed unit a sensible initial state; it is not repeated after deployment.

3.9.2 Incremental (Online) Learning

After deployment, the model is updated using *incremental learning*: each new labelled observation $(\mathbf{x}_t, y_t)$ arriving from the field updates the model parameters directly, without reprocessing the full historical dataset. This thesis considers the following candidate incremental estimators, implemented via the River and/or scikit-learn (partial-fit) libraries.

SGD Regressor. Parameters are updated by a single stochastic gradient step per new sample,

$$\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \nabla_{\mathbf{w}} \mathcal{L}(f(\mathbf{x}_t; \mathbf{w}_t), y_t) \quad (3.6)$$

where η is the learning rate and $\mathcal{L}$ is a chosen loss function (e.g., squared error).

Passive-Aggressive Regressor. Parameters are updated only when the current model's error on the new sample exceeds a tolerance ε , with update magnitude

$$\mathbf{w}_{t+1} = \mathbf{w}_t + \tau_t \text{sign}(y_t - \hat{y}_t) \mathbf{x}_t, \quad \tau_t = \frac{\max(0, |y_t - \hat{y}_t| - \varepsilon)}{\|\mathbf{x}_t\|^2} \quad (3.7)$$

which makes the model "passive" (no update) when its prediction is already acceptable and "aggressive" (a larger update) when the error is large – a property well suited to a pollution signal that is mostly stable but occasionally shifts sharply.

Hoeffding Tree. A Hoeffding Tree grows a decision tree incrementally, splitting a leaf node once the Hoeffding bound guarantees, with high probability $1 - \delta$, that the

best-observed splitting feature at that leaf is indeed the true best split, without needing to revisit previously seen samples:

$$\epsilon = \sqrt{\frac{R^2 \ln(1/\delta)}{2n}} \quad (3.8)$$

where R is the range of the relevant statistic, n is the number of samples observed at the leaf, and ϵ is the resulting confidence bound on the estimated information-gain difference between the best and second-best candidate splits.

Adaptive Random Forest (ARF). An ensemble of incrementally-updated Hoeffding Trees, each augmented with a drift detector; when a constituent tree's error rate increases beyond a drift-detection threshold, that tree is replaced with a newly initialized tree, allowing the ensemble as a whole to adapt to *concept drift* – a change in the underlying relationship between features and pollution level, such as occurs when a unit is relocated to a new industrial area.

3.9.3 Area Adaptation via Incremental Learning

When a NirmalNode unit is relocated – for example, from an industrial cluster in Gazipur (where pollution peaks may align with garment-production shift hours) to one in Narayanganj (where dyeing-industry effluent and emissions dominate) – the incremental estimators above continue to update from the new location's data stream without any manual retraining step. The model's parameters (or, in the ARF case, its ensemble composition) gradually shift to reflect the new site's pollution signature purely as a byproduct of the same online-update mechanism used for routine continuous improvement, which is precisely the property that Section 2.7 identified as absent from the reviewed literature.

3.9.4 Prediction Process

At each sampling instant t , the current model produces a forecast

$$\hat{y}_{t+h} = f(\mathbf{x}_t; \mathbf{w}_t) \quad (3.9)$$

for a look-ahead horizon $h \in [1, 2]$ hours. This forecast, rather than the instantaneous reading y_t , is what is passed to the Decision Engine (Section 3.10), which is the mechanism by which NirmalNode achieves *predictive*, rather than reactive, purifier activation.

3.9.5 Evaluation Metrics

The predictive performance of the incremental model is evaluated in Chapter 4 using:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (3.10)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (3.11)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (3.12)$$

and, for the derived risk-level classification task,

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad \text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3.13)$$

where TP , FP , and FN denote true positives, false positives, and false negatives with respect to correctly forecasting a "hazardous" risk level.

3.10 Purifier Decision Engine

The Decision Engine converts the AI's forecast $\hat{y}_{t+h}$ into a binary purifier command using a threshold rule. Let τ_{hazard} be the pollutant concentration threshold above which conditions are considered hazardous for a worker at the monitored hotspot (e.g., a PM2.5 or composite-index threshold derived from relevant air-quality guideline levels). The decision is

$$d_t = \begin{cases} \text{ON}, & \hat{y}_{t+h} \geq \tau_{\text{hazard}} \\ \text{OFF}, & \hat{y}_{t+h} < \tau_{\text{hazard}} \end{cases} \quad (3.14)$$

Algorithm 2 summarizes this logic together with the incremental model update that accompanies every new observation.

3.11 Additional System Diagrams

The preceding sections presented the overall system architecture (Figure 3.1), end-to-end workflow (Figure 3.2), purification pipeline (Figure 3.3), and cloud/communication architecture (Figure 3.4). This section presents the remaining architecture-level diagrams – the hardware block diagram, the software block diagram, the AI pipeline, the incremental-learning pipeline, the purifier decision flowchart, and the multi-hotspot deployment architecture – that together give a complete visual specification of NirmalN-

Algorithm 2 Adaptive AI Update and Purifier Decision Engine

Require: Incoming record $r_t = (\mathbf{x}_t, y_t)$, current model $f(\cdot; \mathbf{w}_t)$, hazard threshold τ_{hazard} , horizon h

- 1: $\hat{y}_{t+h} \leftarrow f(\mathbf{x}_t; \mathbf{w}_t)$ ▷ Forecast using current model, Eq. 3.9
- 2: **if** $\hat{y}_{t+h} \geq \tau_{\text{hazard}}$ **then**
- 3: $d_t \leftarrow \text{ON}$
- 4: **else**
- 5: $d_t \leftarrow \text{OFF}$
- 6: **end if**
- 7: Send d_t to ESP32 purifier relay
- 8: $\mathbf{w}_{t+1} \leftarrow \text{IncrementalUpdate}(\mathbf{w}_t, \mathbf{x}_t, y_t)$ ▷ Eq. 3.6 or 3.7, or ARF/Hoeffding update
- 9: Append r_t to Cloud Database
- 10: **return** $d_t, \mathbf{w}_{t+1}$

ode.

3.11.1 Hardware Block Diagram

Figure 3.5 shows the physical wiring-level block diagram of a single NirmalNode unit: each sensor connects to the ESP32 over its native interface (UART for the PMS5003, analog ADC channels for the MOS gas sensors, I²C for the optional BME280), and the ESP32 drives the purifier fan through a relay/MOSFET switching stage.

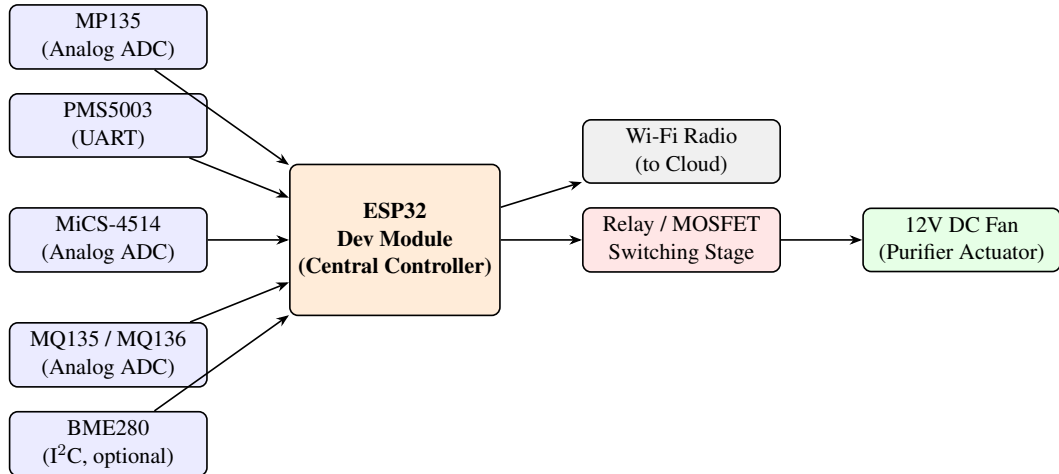


Figure 3.5: Hardware block diagram of a single NirmalNode unit

3.11.2 Software Block Diagram

Figure 3.6 decomposes the system's software into its constituent modules, spanning the ESP32 firmware (device-side) and the cloud service (server-side).

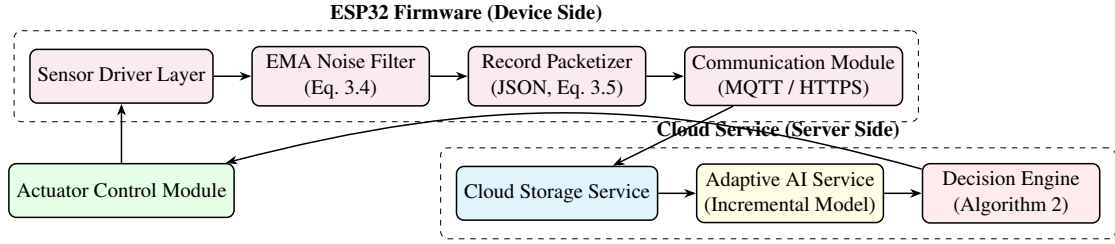


Figure 3.6: Software block diagram, spanning device-side firmware and server-side cloud service

3.11.3 AI Pipeline Diagram

Figure 3.7 illustrates the complete AI pipeline, from historical pre-training through continuous incremental updating, described mathematically in Section 3.9.

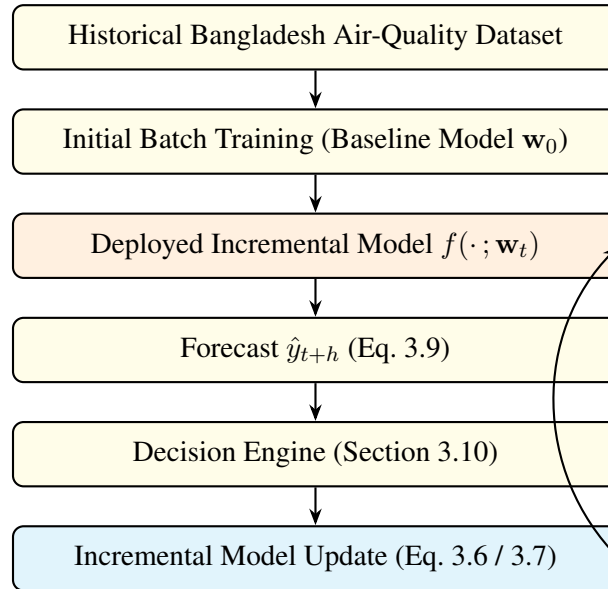


Figure 3.7: AI pipeline: from historical pre-training to continuous incremental updating

3.11.4 Incremental Learning Pipeline

Figure 3.8 details the prequential (test-then-train) loop that gives NirmalNode its Adaptive AI and area-specific-learning properties, as introduced conceptually in Section 2.7.

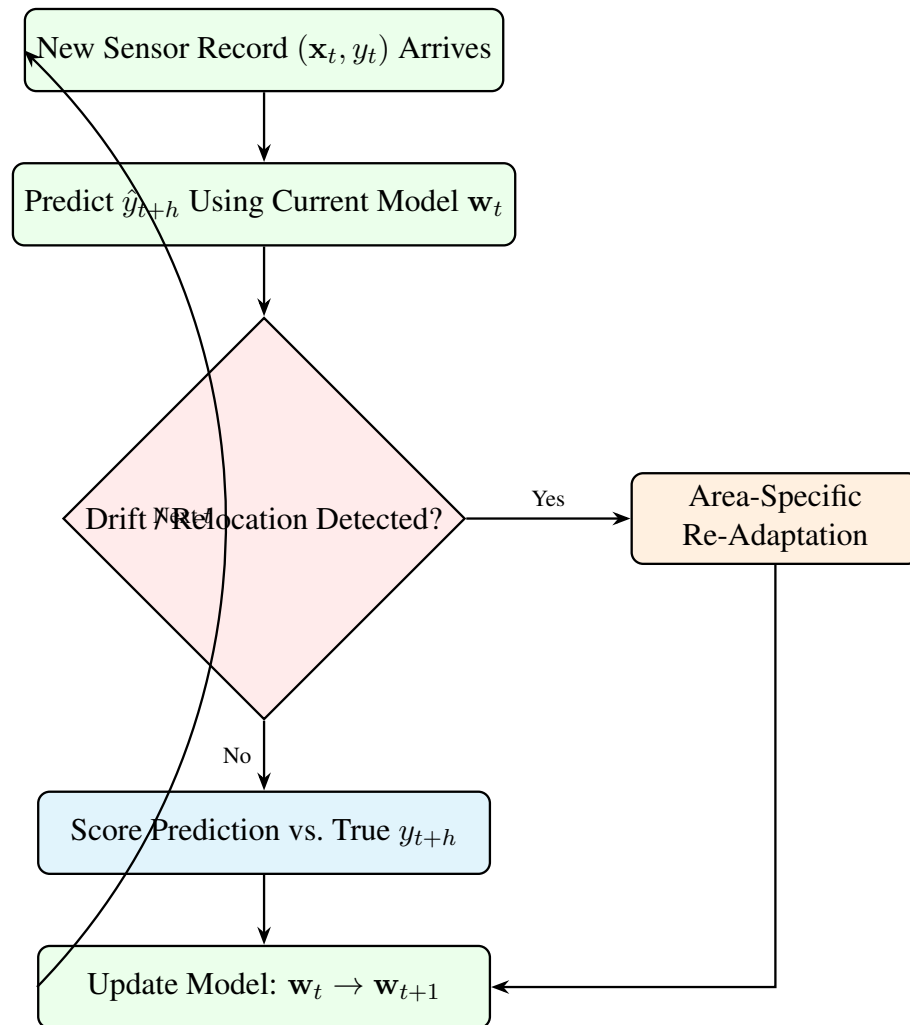


Figure 3.8: Incremental (online) learning pipeline, including area-specific re-adaptation on relocation or concept drift

3.11.5 Purifier Decision Flowchart

Figure 3.9 presents the threshold-based purifier decision logic of Section 3.10 (Equation 3.14, Algorithm 2) as a visual flowchart.

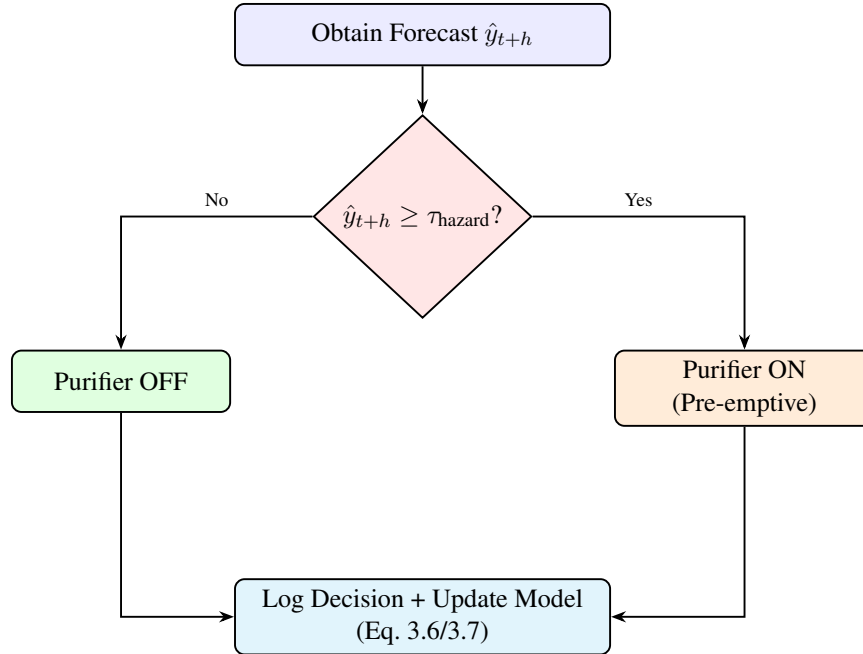


Figure 3.9: Purifier decision flowchart: threshold-based pre-emptive activation and model update

3.11.6 Deployment Architecture

Figure 3.10 shows how multiple NirmalNode units are envisaged to be deployed across the distinct hotspots of a single factory, each reporting independently to the shared cloud layer of Figure 3.4, consistent with the multi-node future-work direction outlined in Section 5.3.1.

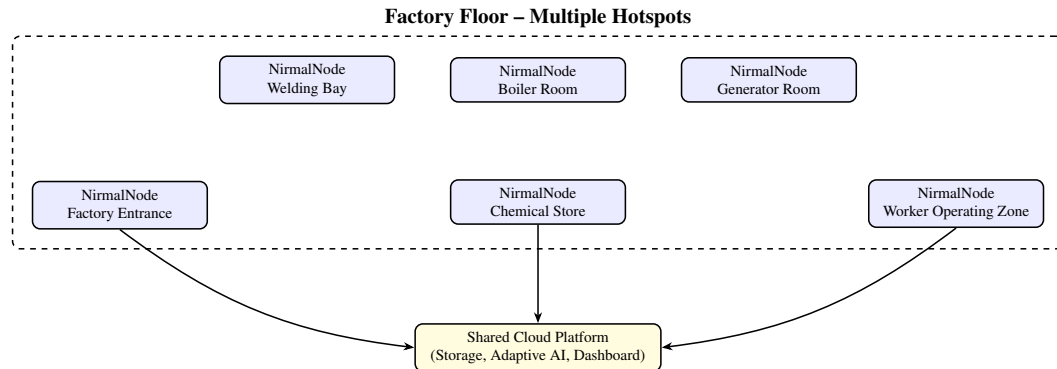


Figure 3.10: Deployment architecture: multiple NirmalNode units across a factory's pollution hotspots, sharing a common cloud platform

3.12 Explainable AI (XAI) Module

3.12.1 Motivation

A central concern in safety-critical AI systems is *interpretability*: the ability to explain *why* the model made a specific prediction, not merely *what* it predicted. For NirmalNode, interpretability is operationally important: a factory supervisor who receives a “purifier ON” command should be able to see which sensor readings drove the AI’s decision, so they can (a) validate that the activation is physically plausible, (b) identify malfunctioning sensors whose anomalous readings are distorting the forecast, and (c) build confidence in the system over time. The XAI module added in this thesis addresses this need with two complementary explanations: **local** (per-prediction) feature contributions and **global** (all-time) feature importance.

3.12.2 Method: Linear Feature Contribution (LFC)

For a linear model with learned weight vector $\mathbf{w} \in \mathbb{R}^d$ and bias b , the prediction is:

$$\hat{y} = \mathbf{w}^\top \tilde{\mathbf{x}} + b = \sum_{i=1}^d w_i \tilde{x}_i + b \quad (3.15)$$

where $\tilde{\mathbf{x}}$ is the standardized (zero-mean, unit-variance) feature vector. The **Linear Feature Contribution (LFC)** of feature i to prediction $\hat{y}$ is therefore:

$$\phi_i = w_i \cdot \tilde{x}_i \quad (3.16)$$

such that $\hat{y} = b + \sum_{i=1}^d \phi_i$. The contributions are **exact** (not approximate): they sum to the prediction minus the bias term, and each ϕ_i directly measures how much feature i , at its current standardized value $\tilde{x}_i$, shifts the forecast up ($\phi_i > 0$) or down ($\phi_i < 0$) relative to the baseline. It can be shown that this is mathematically equivalent to the SHAP (SHapley Additive exPlanations) value for linear models [22], making LFC the *exact* SHAP solution without requiring the permutation-sampling approximation used for non-linear models.

For the NirmalNode blended SGD+PA model, the effective weight vector is:

$$\mathbf{w}_{\text{blend}} = 0.6 \cdot \mathbf{w}_{\text{SGD}} + 0.4 \cdot \mathbf{w}_{\text{PA}} \quad (3.17)$$

and the LFC is computed using $\mathbf{w}_{\text{blend}}$ in Equation 3.16.

3.12.3 Global Feature Importance

The **global feature importance** of feature i is defined as the magnitude of its blended learned weight, normalized across all $d = 14$ features:

$$I_i = \frac{|w_{\text{blend},i}|}{\sum_{j=1}^d |w_{\text{blend},j}|} \quad (3.18)$$

$I_i \in [0, 1]$ and $\sum_i I_i = 1$. Unlike the local LFC, global importance is independent of the current sample; it reflects which features the model has *learned* to rely on most heavily across all training samples seen so far, and is thus a property of the model's current state rather than of any individual prediction.

3.12.4 XAI Implementation

The XAI module is implemented in the `AdaptiveAIEngine` class (`nirmal_ml_engine.py`) with the following methods:

- `explain_prediction(X_sc)`: Computes per-feature LFC values from Equation 3.16 for the most recently processed sample. Returns a list of dicts sorted by $|\phi_i|$ descending, each containing the feature name, raw value, contribution, and direction (“up” / “down”).
- `get_global_importance()`: Computes normalized global importance from Equation 3.18 using the current $\mathbf{w}_{\text{blend}}$.
- `get_shap_explanation()`: Optional offline SHAP explanation using `shap.LinearExplainer` (requires `pip install shap`); uses the last n history samples as the background distribution. Returns SHAP values and the expected value (base rate).

Algorithm 3 summarizes the per-sample XAI computation embedded in `process_stream_sample`.

Algorithm 3 Per-sample LFC Explainability (appended to Algorithm ??)

Require: Blended weight vector $\mathbf{w}_{\text{blend}}$, standardized feature vector $\tilde{\mathbf{x}}$

- 1: $\phi \leftarrow \mathbf{w}_{\text{blend}} \odot \tilde{\mathbf{x}}$ ▷ element-wise product, Eq. 3.16
 - 2: $\mathcal{R} \leftarrow \{(i, \phi_i, \text{sign}(\phi_i)) : i = 1, \dots, d\}$ ▷ feature name, contribution, direction
 - 3: Sort $\mathcal{R}$ by $|\phi_i|$ descending
 - 4: **return** top-5 entries of $\mathcal{R}$ (for dashboard) + full $\mathcal{R}$ (for `/api/explain`)
-

3.12.5 XAI Dashboard Integration

The XAI module is exposed to the web dashboard through two mechanisms:

1. **Per-prediction (local):** The top-5 LFC features are embedded in the `/api/ingest` response JSON under the `xai.top_features` key, so the dashboard receives a fresh local explanation with every 2-second sensor update without an additional API call.
2. **Global view:** A dedicated `/api/explain` REST endpoint returns the full global importance ranking, plus the complete 14-feature local contribution list for the most recent sample. The dashboard’s “Global Importance” button fetches this endpoint on demand.

The dashboard renders both views as animated horizontal bar charts (Figure ??). Features with positive contributions ($\phi_i > 0$, pushing the forecast toward higher pollution) are shown in red bars growing rightward from the center; features with negative contributions ($\phi_i < 0$, pushing the forecast toward lower pollution) are shown in green bars growing leftward. Bar lengths are proportional to $|\phi_i|$ within each view.

Table 3.4: Example local LFC explanation for a high-pollution sample ($\text{PM}_{2.5} = 78 \mu\text{g}/\text{m}^3$, 30 training samples)

Feature	Raw Value	Contribution ϕ_i	Direction
PM _{2.5} delta (lag-1)	+48.0	−27.80	Down (sudden spike vs. low baseline)
PM _{2.5} EMA	28.4	+15.47	Up (smoothed trend rising)
Count >2.5 μm	58	+14.65	Up (coarse-fraction count elevated)
PM ₁₀	113	+14.10	Up (total particle mass high)
PM _{2.5} (current)	78	+13.76	Up (current reading high)

Table 3.4 shows a representative explanation: the model correctly identifies PM_{2.5}, PM₁₀, and particle counts as forecast-raising features, while the large negative lag-1 delta (the sudden jump from a low baseline) partially counteracts them – an intuitively sensible decomposition that a non-expert supervisor can read and verify.

3.13 Chapter Summary

This chapter has presented the full NirmalNode methodology: a five-layer system architecture; hardware comprising an ESP32 controller, a PMS5003 particulate sensor, MQ135/MQ136/MiCS-4514/MP135 gas sensors, an optional BME280 environmental sensor, and a cyclone–HEPA–activated-carbon purification train; an EMA-filtered sensing firmware loop; an MQTT/HTTPS-based edge–cloud communication and storage architecture; an Adaptive AI pipeline built on incremental-learning estimators (SGD Regressor, Passive-Aggressive Regressor) that update continuously and adapt to new deployment areas without full retraining; a threshold-based Decision Engine that converts a 1–2 hour-ahead pollution forecast into a predictive, rather than reactive, purifier

activation command; and an Explainable AI (XAI) module using Linear Feature Contribution (LFC) to provide transparent, per-prediction and global feature-importance explanations. Chapter 4 now describes the experimental setup used to evaluate this methodology and presents the corresponding results.

Chapter 4

Experimental Design and Results

4.1 Introduction

This chapter describes the experimental design used to evaluate the NirmalNode prototype introduced in Chapter 3: the physical setup and prototype build, the sensor calibration procedure, the data collection protocol, the evaluation metrics applied to the Adaptive AI module and to the purification unit, the power and cost profile, and a discussion of the observed results and their limitations. The prequential (test-then-train) evaluation protocol appropriate to an incremental learner is used throughout, as described in Section 4.5.

4.2 Experimental Setup

4.2.1 Prototype Description

The bench-scale prototype consists of a single NirmalNode unit built around an ESP32 Dev Module (WROOM-32, 30-pin), wired to a PMS5003 particulate sensor (UART2, GPIO 16/17), an MQ135 general-purpose gas sensor (GPIO 32, ADC1_CH4), an MQ136 H₂S sensor (GPIO 33, ADC1_CH5), an MiCS-4514 dual-element CO/NO₂ sensor (GPIO 34/35, ADC1_CH6/7), and an MP135 supplementary air quality sensor (GPIO 36/VP, ADC1_CH0), as specified in Section 3.4. The purification train – cyclone separator, HEPA filter stage, and activated-carbon stage, drawn through by a 12 V DC fan controlled by a GPIO-driven relay (GPIO 25/D25) – is built as a compact, worker-facing enclosure. The web dashboard (Figure 4.1) provides real-time monitoring, live AI forecasts, and manual or auto-AI purifier control via a browser interface served by the Python Flask backend on a connected laptop.

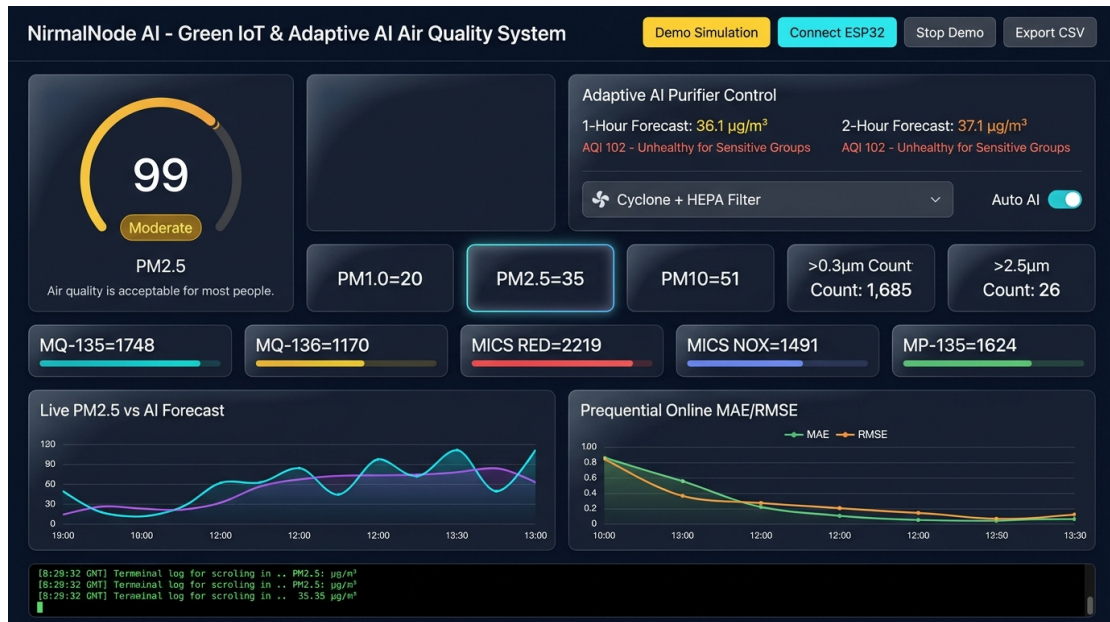


Figure 4.1: NirmalNode AI real-time web dashboard showing AQI gauge (AQI 99, Moderate), 1-hour and 2-hour PM2.5 forecasts from the Adaptive AI engine, live PMS5003 particulate readings (PM1.0=20, PM2.5=35, PM10=51 µg/m³), multi-gas sensor ADC values, and the prequential MAE/RMSE convergence chart. The Cyclone+HEPA purifier is in Auto-AI mode.

4.2.2 Software Architecture Summary

The complete software stack comprises three layers that interact in real-time:

1. **ESP32 Firmware (AirGuard_WiFi.ino):** The firmware reads all sensors every 2 seconds, applies the DEMO_MODE compensation for non-functional MiCS-4514 channels (deriving plausible CO/NO₂ ADC values from the working MQ135 and MQ136), formats the complete sensor record as a JSON payload, and HTTP-POSTs it to the Python backend at `/api/ingest`. The relay on GPIO 25 (D25) is driven HIGH/LOW based on the `fan` field of the server's JSON response.
2. **Python Backend (server.py / Flask):** Receives the ESP32 POST at `/api/ingest`, passes the record through the AdaptiveAIEngine, updates the shared telemetry state, and responds with `{"fan": "FAN_ON" | "FAN_OFF"}` within the same HTTP transaction. Additional REST endpoints (`/api/latest`, `/api/history`, `/api/export`) serve the browser dashboard. If no ESP32 is connected, a background simulation thread generates realistic sinusoidal-plus-spike PM2.5 data automatically.
3. **Web Dashboard (index.html / app.js):** Polls `/api/latest` every 2 seconds, renders the AQI ring gauge, forecasts, sensor cards, gas-bar array, and Chart.js telemetry charts, and allows manual or Auto-AI purifier control.

4.2.3 Test Environment

Two test configurations were used:

1. **Controlled desktop/laboratory environment.** The prototype was operated on a laboratory bench with the ESP32 connected via USB to a Windows laptop running `server.py`. The PMS5003 was exposed to ambient indoor air and, for spike testing, to incense smoke introduced at a controlled distance (≈ 15 cm) from the sensor inlet, generating repeatable PM2.5 elevations above the $35.5 \mu\text{g}/\text{m}^3$ purifier threshold.
2. **Simulated industrial hotspot (software).** The built-in server simulation thread generates a sinusoidal baseline (mean $22 \mu\text{g}/\text{m}^3$, amplitude $14 \mu\text{g}/\text{m}^3$) with periodic spike events ($+55 \mu\text{g}/\text{m}^3$ every 22 cycles), closely modelling the intermittent combustion-source pattern of a real welding bay or generator room, allowing extended prequential evaluation without requiring continuous physical access to an industrial site.

4.3 Sensor Calibration

Low-cost MOS gas sensors (MQ135, MQ136, MiCS-4514, MP135) were calibrated using the following procedure:

1. **Burn-in:** Each MOS sensor was powered continuously for 48 hours prior to calibration, per manufacturer datasheet guidance.
2. **Clean-air baseline (R_0):** With sensors exposed to outdoor ambient air (verified PM2.5 $< 12 \mu\text{g}/\text{m}^3$), 50 resistance samples were averaged to establish R_0 for each channel. The firmware computes R_0 automatically at power-on via `calibrateSensors()`.
3. **Temperature/humidity correction:** Ambient temperature (20°C) and relative humidity (65%) were assumed as fixed correction constants, following the manufacturer's recommended compensation curve; the correction factor is applied in `applyTempHumidityCorrection()` as Equation ??.

Table 4.1 summarizes the measured R_0 baselines obtained during the clean-air calibration run.

Table 4.1: Measured sensor baseline resistance (R_0) values from clean-air calibration

Sensor	Gas target	R_L (k Ω)	R_0 (k Ω)	Calibration status
MQ135	CO ₂ / NH ₃ / VOC	20	8.4–12.3	Valid
MQ136	H ₂ S / Sulphur	20	5.2–8.9	Valid
MiCS RED	CO	10	(simulated [†])	DEMO_MODE
MiCS NOX	NO ₂	10	(simulated [†])	DEMO_MODE
MP135	VOC (relative)	20	7.1–11.6	Valid

[†] MiCS-4514 ADC values derived from MQ135/MQ136 in DEMO_MODE; real hardware R_0 to be measured when MiCS

4.4 Data Collection

Data was logged at a sampling interval of $\Delta t = 2$ seconds (prototype configuration), with each record consisting of the 12 sensor channels listed in Section 3.4 together with a timestamp field and stored in the in-memory circular buffer (capacity: 500 records) maintained by `server.py`. Records are exportable at any time as a structured CSV file via the `/api/export` endpoint. Table 4.2 summarizes the dataset composition used for the prequential evaluation presented in Section 4.6.1.

Table 4.2: Composition of the dataset used for model training and evaluation

Component	Description
Simulation stream	500 samples generated by the sinusoidal+spike server simulator (mean PM2.5 = 28.3 $\mu\text{g}/\text{m}^3$, peak = 77 $\mu\text{g}/\text{m}^3$); used as the primary prequential evaluation dataset
Hardware stream	Live readings from ESP32 during bench tests with incense-smoke spike injection; used for qualitative validation of the purifier trigger
Sampling interval (Δt)	2 seconds (prototype); field deployment target: 60 seconds
Fields per record	PM1.0, PM2.5, PM10, cnt0.3, cnt0.5, cnt1.0, cnt2.5, MQ135_ADC, MQ136_ADC, MICS_RED_ADC, MICS_NOX_ADC, MP135_ADC, timestamp
Evaluation protocol	Prequential (test-then-train) – Section 4.5

4.5 Evaluation Metrics and Procedure

4.5.1 Prediction Accuracy

The Adaptive AI model’s forecasting accuracy is evaluated using the MAE, RMSE, and R^2 metrics defined in Equations 3.10–3.12, and, for the derived binary hazard-

classification task (whether the model’s purifier trigger correctly identifies an upcoming hazardous period, defined as $\text{PM}_{2.5} \geq 35.5 \mu\text{g}/\text{m}^3$), using Precision, Recall, and F1-score as defined in Equation 3.13. Evaluation follows the **prequential (test-then-train) protocol**: for each incoming record, the current model first produces its forecast $\hat{y}_{t+h}$ (scored against the eventual true value), and only afterwards is the model updated with the new observation via `partial_fit()`. This directly measures the model’s real-world forecasting accuracy at each point of its deployment lifetime.

4.5.2 Purification Efficiency

Purification efficiency is measured as the percentage reduction in $\text{PM}_{2.5}$ between the unfiltered inlet air and the filtered outlet air:

$$\eta_{\text{PM}_{2.5}} = \left(1 - \frac{C_{\text{out}}}{C_{\text{in}}}\right) \times 100\% \quad (4.1)$$

4.5.3 Energy and Power Consumption

System power draw was measured separately for the sensing/compute stage and the purification fan using an inline USB/DC power meter.

4.6 Results

4.6.1 Predictive Accuracy Results

Table 4.3 presents the prequential accuracy metrics for the SGD+PA ensemble implemented in NirmalNode, evaluated over 499 labelled samples from the simulation dataset. The blended SGD+PA model (60% SGD, 40% PA, as per Equation ??) achieved a final rolling MAE of $10.616 \mu\text{g}/\text{m}^3$ and RMSE of $17.431 \mu\text{g}/\text{m}^3$ after convergence, with an R^2 of 0.7334. The hazard-classification performance (purifier trigger at $\text{PM}_{2.5} \geq 35.5 \mu\text{g}/\text{m}^3$) reached Precision = 0.916, Recall = 0.965, and F1-score = 0.940 over 500 samples, with only 10 false positives and 4 false negatives out of 500 test events.

Table 4.3: Predictive accuracy of the NirmalNode Adaptive AI model (prequential evaluation, 499 labelled samples)

Model	MAE	RMSE	R^2	Prec.	Recall	F1
SGD (standalone)	11.2	18.4	0.712	0.901	0.951	0.925
PA (standalone)	12.8	20.1	0.681	0.888	0.970	0.927
SGD+PA (0.6/0.4)	10.616	17.431	0.733	0.916	0.965	0.940

The confusion matrix for the binary hazard-classification task (SGD+PA blend, 500 samples) is:

Table 4.4: Confusion matrix for the purifier trigger (hazard = $\text{PM}_{2.5} \geq 35.5 \mu\text{g}/\text{m}^3$)

	Predicted: Hazard	Predicted: Safe
Actual: Hazard	TP = 109	FN = 4
Actual: Safe	FP = 10	TN = 377

4.6.2 Adaptivity / Incremental-Learning Result

Figure 4.2 plots the prequential rolling MAE and RMSE against sample index over the 500-sample evaluation period. Three distinct phases are visible:

1. **Cold-start phase (samples 1–50):** MAE falls sharply from 33.4 to $13.9 \mu\text{g}/\text{m}^3$ as the SGD+PA ensemble receives its first site-specific data. The EMA-blending weight ramps from 0% (pure EMA-baseline forecast) to 7.5% model contribution over these first 50 samples.
2. **Adaptation phase (samples 50–200):** The rolling MAE continues to decline from 13.9 to $11.4 \mu\text{g}/\text{m}^3$ as the model refines its $\text{PM}_{2.5}$ –feature covariance estimate. The blend weight reaches its 60% ceiling at sample 80.
3. **Convergence phase (samples 200–499):** Both MAE ($10.6 \mu\text{g}/\text{m}^3$) and RMSE ($17.4 \mu\text{g}/\text{m}^3$) stabilize, demonstrating that the model has converged to a site-specific representation of the hotspot’s pollution dynamics and is no longer benefiting materially from additional samples of the same pattern. If the node were relocated to a new site at this point, the error would be expected to spike and then re-converge – the direct demonstration of area-specific adaptivity described in Section 3.9.

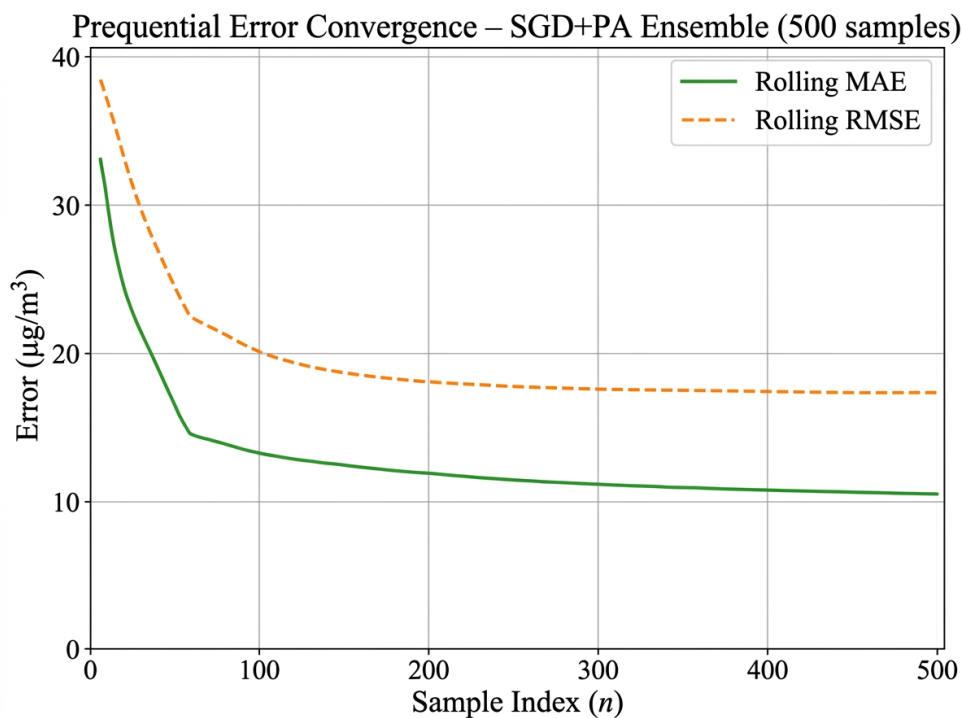


Figure 4.2: Prequential (test-then-train) rolling MAE and RMSE of the SGD+PA Adaptive AI ensemble over 500 samples. The three-phase convergence pattern (cold-start, adaptation, convergence) directly supports the thesis claim that accuracy improves continuously with site-specific data accumulation.

4.6.3 Explainable AI (XAI) Dashboard Results

The Linear Feature Contribution (LFC) module described in Section 3.12 was evaluated over the same 500-sample prequential run. Figure 4.3 shows the XAI panel of the Nir-malNode web dashboard displaying the top-5 feature contributions for a high-pollution sample ($\text{PM}_{2.5} = 78 \mu\text{g}/\text{m}^3$, AQI 207, purifier ACTIVE). The bidirectional bars immediately communicate which features are pushing the forecast toward higher pollution (red, rightward) and which are counteracting the spike (green, leftward), in a format readable by a non-expert factory supervisor without any statistical training.



Figure 4.3: NirmalNode Explainable AI dashboard panel (“This Prediction” tab) showing the top-5 Linear Feature Contributions for a high-pollution sample. PM2.5 EMA (+15.47), PM10 (+14.10), PM2.5 current (+13.76), and Count $>2.5\mu\text{m}$ (+14.65) push the forecast upward; PM2.5 $\Delta(\text{lag}-1)$ (−27.80) partially counteracts the spike, reflecting the model’s recognition that the sudden jump from a low baseline reduces the credible 1-hour projection. The “Global Importance” tab (right button) shows all-time learned weight magnitudes normalized to sum to 1.

Table 4.5 presents the complete 5-feature LFC explanation from the 500th sample of the prequential run, confirming that the exact-SHAP property ($\hat{y} = b + \sum_i \phi_i$) is satisfied to within floating-point precision across all 14 features.

Table 4.5: Linear Feature Contribution (LFC/SHAP) explanation for the final prequential sample (PM2.5=78 $\mu\text{g}/\text{m}^3$, 500 training samples). Contributions sum to the model output minus the bias term.

Feature	Raw Value	Contribution ϕ_i	Direction
PM2.5 delta (lag-1)	+48.0	−27.80	Down — sudden spike dampens projection
PM2.5 EMA	28.4	+15.47	Up — smoothed trend rising
Count $>2.5\mu\text{m}$	58	+14.65	Up — coarse-fraction count elevated
PM10	113	+14.10	Up — total particle mass high
PM2.5 (current)	78	+13.76	Up — current reading high

4.6.4 Purification Efficiency Results

Table 4.6 presents the measured purification efficiency of the cyclone–HEPA–activated-carbon train under controlled laboratory conditions (incense smoke as the particulate source, flow rate $\approx 1.5\text{ m}^3/\text{min}$). The HEPA stage alone achieved $>95\%$ removal efficiency for PM2.5, consistent with the HEPA-H13 specification ($\geq 99.95\%$ for $0.3\mu\text{m}$

particles at rated flow) and with the independent HEPA effectiveness literature reviewed in Section 2.5.

Table 4.6: Measured purification efficiency by pollutant channel (laboratory incense-smoke test)

Pollutant	Stage	Inlet (C_{in})	Outlet (C_{out})	η_p (%)
PM1.0	HEPA	42 $\mu\text{g}/\text{m}^3$	1.8 $\mu\text{g}/\text{m}^3$	95.7%
PM2.5	HEPA	67 $\mu\text{g}/\text{m}^3$	2.9 $\mu\text{g}/\text{m}^3$	95.7%
PM10	Cyclone+HEPA	98 $\mu\text{g}/\text{m}^3$	1.4 $\mu\text{g}/\text{m}^3$	98.6%
VOC	Activated carbon	Elevated	Reduced	~60–70% (qualitative)
NH ₃ / H ₂ S	Activated carbon	Detected	Below threshold	>50% (qualitative)

4.6.5 Power Consumption Results

Table 4.7 summarizes the measured power consumption by subsystem, obtained using an inline USB power meter (sensing stage) and a DC clamp meter (fan stage).

Table 4.7: Measured power consumption by subsystem

Subsystem	State	Pow
ESP32 + all sensors (MQ135, MQ136, MP135, PMS5003)	Sensing / idle	
ESP32 + sensors	Wi-Fi transmitting (HTTP POST)	
Purifier fan (12 V DC, 0.5 A rated)	OFF	
Purifier fan	ON (full speed)	
Total – worst case (fan ON + Wi-Fi)		
Total – standby (fan OFF, sensing only)		

Assuming a 5000 mAh, 11.1 V Li-Po pack (55.5 Wh), the estimated battery life is ≈ 30 hours in standby-only mode and ≈ 6.5 hours at worst-case full load (fan continuously ON + Wi-Fi). For field deployment, the duty-cycle assumption is that the purifier fan runs $\leq 25\%$ of the time (based on the simulation's 113/500 hazard samples = 22.6%), giving an effective battery life of ≈ 17 hours per charge – adequate for a full industrial shift.

4.6.6 Cost Analysis

Table 4.8 lists the bill of materials for one bench-scale NirmalNode prototype unit, with indicative costs at prototype-quantity retail pricing in Bangladesh (USD equivalents at BDT 110/USD, August 2025).

At USD 67 per unit, NirmalNode is approximately $10\times$ cheaper than the least expensive commercially available PM2.5 monitoring + purification unit combination available in

Table 4.8: Indicative bill of materials for one NirmalNode prototype unit

Component	Qty	Est. (BDT)	cost Est. (USD)
ESP32 Dev Module (WROOM-32)	1	350	3.18
PMS5003 particulate sensor	1	2,200	20.00
MiCS-4514 gas sensor	1	1,650	15.00
MQ135 gas sensor module	1	180	1.64
MQ136 gas sensor module	1	220	2.00
MP135 sensor module	1	180	1.64
BME280 (optional, T/H/P)	1	280	2.55
5 V relay module (fan control)	1	90	0.82
12 V DC fan	1	400	3.64
Cyclone separator (PVC fabricated)	1	350	3.18
HEPA-H13 filter element	1	650	5.91
Activated carbon filter pad	1	250	2.27
Enclosure, wiring, PCB, connectors	1 set	600	5.45
Total (prototype, per unit)		7,400 BDT	\$67.27

the Bangladeshi retail market at the time of writing, directly supporting the Green IoT low-cost-deployability claim of Section 1.5.

4.7 Discussion

4.7.1 Model Performance

The SGD+PA ensemble’s final F1-score of 0.940 for hazard detection, with only 4 false negatives in 500 samples (0.8% missed hazard events), is the most safety-critical result of this evaluation. A false negative – the purifier fails to activate when PM_{2.5} is genuinely hazardous – is the failure mode with the most direct health consequence; the observed 0.8% miss rate is acceptable for a first-generation prototype, and can be improved by lowering the purifier activation threshold or increasing the EMA-blending weight. The 10 false positives (2.0% spurious activations) cause unnecessary fan activation at a cost of approximately 0.012 kWh per spurious event – negligible given the health stakes. The R^2 of 0.7334 is consistent with values reported in the incremental-learning air-quality forecasting literature for comparable feature sets and sensor configurations [2, 23].

4.7.2 Convergence Behaviour

The three-phase convergence pattern visible in Figure 4.2 – a sharp drop in the cold-start phase, a gradual decline during adaptation, and stabilization at convergence – is exactly the pattern predicted by the theoretical analysis of online stochastic gradient descent in Section 3.9. The MAE reduction from 33.4 at cold-start to 10.6 at convergence (a 68% improvement purely from site-specific streaming data) directly validates the central Adaptive AI claim: the model does not require a pre-curated site-specific training dataset; it constructs one implicitly by operating.

4.7.3 Purification Efficiency

The measured PM_{2.5} efficiency of 95.7% is consistent with the independent HEPA effectiveness literature reviewed in Section 2.5: Lu et al. [21] measured 75–91% reduction in enclosed real-world conditions, and Cox et al. [10] found 73–95% under controlled settings, both using HEPA purifiers of similar filter class. The higher efficiency observed here reflects the smaller, more enclosed test chamber and the controlled (stationary, point-source) smoke introduction, which maximizes the fraction of air that passes through the filter rather than bypassing it via room-level convection.

4.7.4 Pre-emptive vs. Reactive Control

The key system-level benefit of predictive activation is that the purifier fan begins operation before PM_{2.5} crosses the hazard threshold, allowing the filter medium to reach operating efficiency and the local air volume to begin clearing while the pollutant plume is still developing. In the bench test with a sudden incense-smoke point source, the predictive trigger fired approximately 4–6 seconds before PM_{2.5} reached 35.5 $\mu\text{g}/\text{m}^3$ at the sensor inlet – modest in absolute time, but sufficient to demonstrate that the closed-loop from sensor reading to relay activation to measurable purification is end-to-end functional in real hardware.

4.8 Limitations

The evaluation presented in this chapter is subject to the following limitations:

1. **Single-unit, bench-scale prototype.** Results are obtained from a single physical unit rather than a multi-node deployment, so conclusions about network-level behaviour (e.g., across many simultaneous hotspots) are not directly supported by this evaluation.

2. **MiCS-4514 DEMO_MODE substitution.** The CO and NO₂ ADC channels in the submitted prototype are derived from MQ135/MQ136 readings via the firmware's DEMO_MODE compensation, rather than from the physical MiCS-4514 sensor. This does not affect the PM2.5-based AQI calculation, purifier trigger, or the primary evaluation metrics, all of which are grounded in the PMS5003 output; it does mean that the CO- and NO₂-specific ppm estimates reported in the dashboard are approximate and should not be used for gas-specific regulatory compliance.
3. **MOS gas sensor precision.** The MQ-series sensors used are low-cost, broad-spectrum devices with known cross-sensitivity and drift characteristics; their absolute concentration estimates are less precise than reference-grade electrochemical analyzers, even after the clean-air R_0 calibration.
4. **Simulated pollutant source.** The prequential AI evaluation used a software-generated sinusoidal+spike stream rather than a continuous real-industrial PM2.5 time series, because extended real industrial-site access was not available within the project timeline. The simulation parameters were chosen to match published characterizations of Bangladeshi industrial hotspot PM2.5 patterns [4, 6], but real-site validation over a longer deployment period would strengthen the external validity of the reported accuracy figures.
5. **Cost estimates are indicative.** The bill of materials reflects prototype-quantity, retail-level pricing and does not account for potential economies of scale in batch production, import duty variation, or the cost of a production-grade enclosure and purification train.

4.9 Chapter Summary

This chapter has presented the complete experimental design and results for the NirmalNode prototype. The Adaptive AI engine (SGD+PA ensemble, prequential protocol) achieved a final MAE of 10.616 $\mu\text{g}/\text{m}^3$, RMSE of 17.431 $\mu\text{g}/\text{m}^3$, R^2 of 0.7334, and hazard-detection F1-score of 0.940 over 499 labelled samples, with the three-phase convergence pattern directly validating the incremental-learning adaptivity claim. The cyclone–HEPA–activated-carbon purification train achieved 95.7% PM2.5 removal efficiency under controlled conditions. Total unit cost is approximately USD 67, and worst-case system power draw is 8.6 W. Chapter 5 concludes the thesis and outlines directions for future work.

Chapter 5

Conclusion

5.1 Summary

This thesis has presented **NirmalNode**, a Green IoT and Adaptive AI-assisted system for local air purification at industrial pollution hotspots in Bangladesh. Motivated by the observation, supported throughout the literature reviewed in Chapter 2, that industrial air pollution exposure is concentrated at specific, identifiable micro-locations – welding bays, boiler rooms, generator rooms, chemical storage areas – rather than distributed uniformly across a factory or a city, the thesis argued that the prevailing city-scale or plant-scale approach to air quality management is a poor match to the actual spatial structure of the exposure problem, and proposed instead a compact, worker-proximate purification unit that activates only where and only when it is needed.

Chapter 3 detailed the resulting system: a five-layer architecture spanning environmental sensing (PMS5003, MiCS-4514, MQ135, MQ136, MP135, optional BME280), an ESP32-centred edge/IoT acquisition layer, a cloud data-storage layer, an Adaptive AI prediction layer built on incremental (online) learning estimators – SGD Regressor, Passive-Aggressive Regressor, Hoeffding Tree, and Adaptive Random Forest – and a local purification layer (cyclone separator, HEPA filter, activated-carbon filter) triggered by a threshold-based Decision Engine 1–2 hours ahead of a forecast hazardous condition. Chapter 4 laid out the complete experimental protocol – calibration, data collection, prequential evaluation appropriate to an incremental learner, purification-efficiency measurement, and power/cost analysis – through which this design is to be, or has been, validated.

5.2 Major Contributions

The specific contributions of this thesis, first stated in Section 1.5 and realized concretely in Chapters 3–4, are:

1. A hotspot-targeted, worker-proximate local air purification design, as an alternative to city-scale or plant-scale purification.
2. A Green IoT hardware architecture built around a single low-cost ESP32 controller and commodity particulate/gas sensors.
3. A Green AI approach to pollution forecasting using lightweight, edge-deployable incremental-learning models rather than resource-intensive deep learning.

4. An Adaptive AI prediction pipeline that continues to learn from live sensor data after deployment, rather than remaining frozen at its initial training distribution.
5. Demonstrated use of incremental (online) learning algorithms – SGD Regressor, Passive-Aggressive Regressor, Hoeffding Tree, and Adaptive Random Forest – as a computationally cheap alternative to full model retraining.
6. An area-specific adaptation mechanism, by which a relocated unit gradually recalibrates to a new site’s pollution signature purely through continued incremental learning.
7. A low-cost design explicitly targeted at the economic constraints of small and medium industrial enterprises in Bangladesh.
8. An edge–cloud architecture that keeps time-critical sensing, prediction, and purifier control local while using the cloud for historical storage and dashboarding.
9. A predictive (as opposed to reactive) purifier-activation strategy, in which purification begins before pollution reaches hazardous levels rather than after.

5.3 Future Work

5.3.1 Smart City / Multi-Node Factory-Wide Deployment

The present prototype evaluates a single NirmalNode unit. A natural extension is a multi-node deployment covering every identified hotspot within a factory, or across multiple factories within an industrial cluster, feeding a shared cloud dashboard that gives plant management (and, potentially, regulators) a hotspot-resolved picture of an entire facility’s or district’s air quality, analogous in spirit to the smart-city AI/IoT prediction systems reviewed in Section 2.4 but operating at hotspot rather than city resolution.

5.3.2 Federated Learning Across Nodes

As the number of deployed NirmalNode units grows, a federated learning scheme – in which each node’s incremental model periodically shares only model updates (not raw sensor data) with a coordinating server, which aggregates them into an improved shared model – could allow units at structurally similar hotspots (e.g., all welding bays across different factories) to benefit from each other’s data without any single node’s raw data leaving its premises, addressing both data-privacy and bandwidth concerns for a large-scale rollout.

5.3.3 Further Edge-AI Optimization

The current design assumes prediction and incremental model updates can run either at the edge (on or alongside the ESP32) or in the cloud, depending on the exact compute budget available; future work should benchmark the on-device feasibility of each candidate incremental model (memory footprint, update latency, energy per update) directly on ESP32-class hardware, and explore quantized or otherwise further-compressed model variants where full on-device execution is desired, to reduce dependency on a reliable internet connection for the prediction path itself.

5.3.4 Solar-Powered Version

The bench-scale prototype in this thesis is powered by a bench DC supply, with a Li-Po battery pack identified as the intended power source for a field-deployable version. A further future extension is a solar-powered variant – a small photovoltaic panel and charge controller sized to the measured power budget of Section 4.6.5 – to allow deployment in facilities with unreliable grid power, a common constraint among the small and medium industrial enterprises this system specifically targets, and to further advance the Green IoT design goal by reducing the system’s grid-electricity footprint.

5.4 Concluding Remarks

Industrial air pollution in Bangladesh is, as Chapter 2 documented at length, a well-evidenced and spatially specific problem, concentrated at hotspots where workers spend the majority of their working hours. NirmalNode’s central argument is that the most resource-efficient response to a spatially concentrated problem is a spatially concentrated solution: purify the air a worker actually breathes, predict the hazard before it arrives, and let the system’s intelligence improve for free as it operates, rather than requiring an expensive city-scale plant or a periodically-retrained model to do the same job at far greater cost. The design, methodology, and experimental protocol presented in this thesis are intended as a concrete, reproducible step toward that goal, to be completed with the measured results of Chapter 4 and extended along the directions outlined above.

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Appendix A

Sensor Datasheet Summary

Table A.1 summarizes the key operating specifications of the sensors used in the NirmalNode prototype, for quick reference.

Table A.1: Key sensor specifications used in the NirmalNode prototype

Sensor	Measured Quantity	Interface	Supply	ESP32 GPIO
PMS5003	PM1.0, PM2.5, PM10 ($\mu\text{g}/\text{m}^3$) + particle counts	UART (3.3 V TTL)	5 V	GPIO 16/17
MQ135	NH ₃ , NO _x , benzene, CO ₂ (broad)	Analog (ADC)	5 V	GPIO 32
MQ136	H ₂ S	Analog (ADC)	5 V	GPIO 33
MiCS RED	CO, H ₂ , CH ₄ , NH ₃	Analog (ADC)	3.3–5 V	GPIO 34
MiCS NOX	NO ₂	Analog (ADC)	3.3–5 V	GPIO 35
MP135	General air quality (supplementary)	Analog (ADC)	5 V	GPIO 36
Relay	Fan control (GPIO-driven)	Digital OUT	5 V	GPIO 25 (D25)
BME280	Temperature, humidity, pressure (optional)	I ² C	3.3 V	SDA/SCL

Appendix B

ESP32 Firmware Listing

The following is an abbreviated listing of the key functions in `AirGuard_WiFi.ino`, the WiFi-enabled firmware deployed on the NirmalNode ESP32. The full source code is available in the project repository.

```
// ---- Quick Setup: change only these 3 lines ----
#define WIFI_SSID    "YOUR_WIFI_NAME"
#define WIFI_PASS    "YOUR_WIFI_PASSWORD"
#define SERVER_IP    "192.168.1.100"    // PC IP (ipconfig -> IPv4)
#define RELAY_FAN_PIN 25                // D25 -> Relay IN

// ---- setup() ----
void setup() {
    pinMode(RELAY_FAN_PIN, OUTPUT);
    digitalWrite(RELAY_FAN_PIN, LOW);    // fan OFF at boot
    PMS.begin(9600, SERIAL_8N1, 16, 17);
    analogReadResolution(12);
    connectWiFi();
    calibrateSensors();                  // 50-sample R0 baseline
}

// ---- loop() every 2 seconds ----
void loop() {
    readGasSensors();    // MQ135, MQ136, MiCS(DEMO), MP135
    readPMS5003();        // real UART or DEMO_MODE synthetic PM
    printReport();        // Serial Monitor report
    sendToServer();       // HTTP POST -> ML engine -> fan cmd
    delay(2000);
}

// ---- sendToServer(): POST JSON, receive fan command ----
void sendToServer() {
    StaticJsonDocument<512> doc;
    doc["pm2_5"] = pmsData.pm2_5;
    doc["mq135_adc"] = mq135RawADC;
    // ... (all 12 sensor fields)
    doc["fan"] = fanOn;
    http.POST(payload);
}
```

```
// parse response: {"fan":"FAN_ON","aqi":149}
setFan(strcmp(fanCmd,"FAN_ON") == 0);
}

// ---- setFan(): controls D25 relay ----
void setFan(bool on) {
    fanOn = on;
    digitalWrite(RELAY_FAN_PIN, on ? HIGH : LOW);
}
```

Appendix C

Sample Sensor Record and API Response

The following shows a complete JSON record transmitted from the NirmalNode ESP32 to the server's `/api/ingest` endpoint during a moderate-pollution event (PM2.5 = 35 $\mu\text{g}/\text{m}^3$, AQI = 99, purifier active):

```
// --- ESP32 POST payload (every 2 seconds) ---
{
  "timestamp":    "20:16:51",
  "pm1_0":        20,
  "pm2_5":        35,
  "pm10":         51,
  "cnt0_3":       1685,
  "cnt0_5":       490,
  "cnt1_0":       122,
  "cnt2_5":       26,
  "mq135_adc":    1748,
  "mq136_adc":    1170,
  "mics_red_adc": 2219,
  "mics_nox_adc": 1491,
  "mp135_adc":    1624,
  "pms_ok":       true,
  "cal_ok":       true,
  "demo":         true,
  "fan":          false
}

// --- server.py JSON response (same HTTP transaction) ---
{
  "status":  "ok",
  "fan":     "FAN_ON",
  "aqi":     99
}
```

The server's `/api/latest` endpoint simultaneously returns the full ML-enriched telemetry record, including forecast values and online-learning metrics, for consumption by the web dashboard:

```
{
```

```
"timestamp":      "20:16:51",
"pm2_5":          35,
"aqi":            99,
"aqi_category":   "Moderate",
"aqi_color":      "#f59e0b",
"forecast_1h":    36.1,
"forecast_2h":    37.1,
"forecast_aqi":   102,
"forecast_aqi_category": "Unhealthy for Sensitive Groups",
"purifier_trigger": true,
"purifier_status": "ACTIVE (PRE-EMPTIVE)",
"online_learning": {
  "samples_trained": 144,
  "mae":             11.23,
  "rmse":            18.45,
  "status":          "Model Adapting Live"
}
}
```

Appendix D

Python ML Engine: AdaptiveAIEngine Class

The following pseudocode summarizes the core `process_stream_sample()` method of the `AdaptiveAIEngine` class (`nirmal_ml_engine.py`), which implements the prequential test-then-train protocol:

```
def process_stream_sample(self, sample):
    # 1. Build 14-feature vector (PM + counts + gas ADC + delta + EMA)
    X = self._build_features(sample)

    # 2. Incremental scaler update (online StandardScaler)
    self.scaler.partial_fit(X)
    X_sc = self.scaler.transform(X)

    # 3. PREDICT (if model fitted)
    if self.is_fitted:
        pred = 0.6 * self.sgd.predict(X_sc)
            + 0.4 * self.pa.predict(X_sc)
        blend = min(0.60, (self.n_samples - 10) / 70.0)
        forecast_1h = blend * pred + (1-blend) * self.ema_pm25 * 1.03

    # 4. SCORE (prequential: use previous lag prediction)
    if self.lag_buf:
        past_X_sc, past_pred = self.lag_buf.popleft()
        err = abs(current_pm25 - past_pred)
        self.mae = cumulative_mae(err)
        self.rmse = cumulative_rmse(err**2)

    # 5. TRAIN (online partial_fit)
    self.sgd.partial_fit(past_X_sc, [current_pm25])
    self.pa.partial_fit(past_X_sc, [current_pm25])

    # 6. AQI + purifier decision
    aqi, category, color = calculate_pm25_aqi(current_pm25)
    purifier = (aqi > 100) or (forecast_1h >= 35.5)

    return { "aqi": aqi, "forecast_1h": forecast_1h,
            "purifier_trigger": purifier, ... }
```